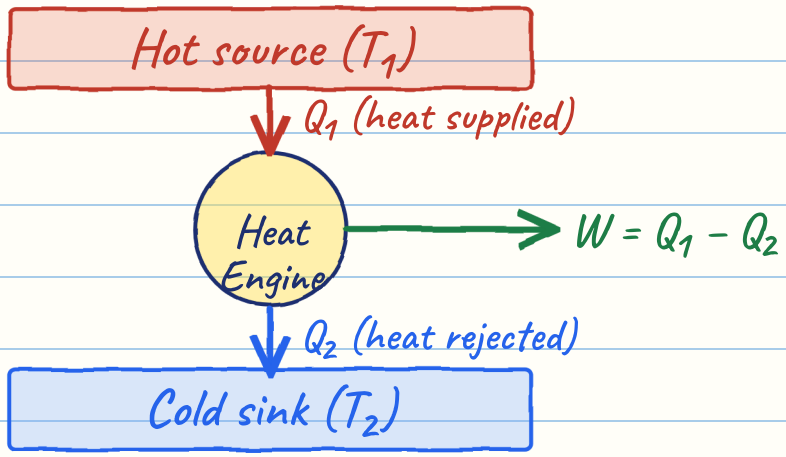
UNIT – I : I.C. ENGINES

I.1 Engine & Heat Engine

- **Engine** : a machine which converts one form of energy into mechanical work (shaft power).
- **Heat engine** : a device that converts heat energy (from burning fuel) into mechanical work continuously, using a working fluid.
- Working fluid receives heat Q_1 from a high temp. source, converts part into work W and rejects remaining heat Q_2 to a low temp. sink.

$\eta = W / Q_1$
$= 1 - Q_2/Q_1$



Thermal efficiency $\eta = W/Q_1 = 1 - Q_2/Q_1$ (always < 100%, Kelvin-Planck statement)

I.2 Classification of Heat Engines

- **External combustion (EC)** engines : fuel burns OUTSIDE the engine cylinder. Eg: steam engine, steam turbine, closed-cycle gas turbine, Stirling engine.
- **Internal combustion (IC)** engines : fuel burns INSIDE the cylinder & the products of combustion themselves act as the working fluid. Eg: petrol engine, diesel engine, gas engine, open-cycle gas turbine.

- Combustion in IC engines is intermittent (reciprocating) while in gas turbines it is continuous.

1.3 EC vs IC Engines

Point	EC engine	IC engine
Combustion	outside cylinder (boiler)	inside the cylinder
Working fluid	steam / air (separate)	products of combustion
Size & weight	bulky, heavy (boiler, condenser)	compact and light
Efficiency	low (15-25%)	high (30-40%)
Starting	slow (need steam raising)	quick starting
Fuel	any fuel, even solid	clean liquid / gas fuel
Power / weight	low	high
Cooling water	heavy (condenser)	less
Uses	power plants	automobiles, pumps, DG sets

1.4 Advantages of IC engines

- Higher overall efficiency & better power-to-weight ratio.
- Compact, light, so suitable for vehicles, aircraft, ships.
- Quick starting, low initial cost for small powers.
- Fuel is easily handled / stored; no boiler or furnace needed.

1.5 Disadvantages

- Cannot use solid fuels; needs costly refined fuels (petrol / diesel).
- Not self starting - needs a starter motor.
- High vibration & noise due to reciprocating parts.
- Exhaust gases cause air pollution (CO, HC, NOx, PM).
- Lower torque at low speeds; needs clutch & gear box.

1.6 Classification of IC Engines

(a) Based on cycle of operation

- **Otto cycle** (constant volume heat addition) - petrol / SI engines.
- **Diesel cycle** (const. pressure heat addition) - slow speed CI engines.
- **Dual cycle** (part at const. V, part at const. p) - high speed diesel engines.

(b) Based on number of strokes per cycle

- **4-stroke** engine : cycle completed in 4 strokes = 2 revolutions (720°).
- **2-stroke** engine : cycle completed in 2 strokes = 1 revolution (360°).

(c) Based on ignition

- **Spark ignition (SI)** - petrol, gas engines; spark plug ignites mixture.
- **Compression ignition (CI)** - diesel; fuel self-ignites in hot compressed air.

(d) Based on fuel used

- Petrol, diesel, gas (CNG, LPG), bi-fuel / dual fuel, alcohol (ethanol), biodiesel.

(e) Based on cylinder arrangement

- **Inline**, **V-type**, **opposed (flat)**, **radial**, **W / X / H** types.
- Vertical or horizontal; single cylinder or multi-cylinder.

(f) Based on cooling : air cooled / water cooled.

(g) Based on valve location : L, T, I (overhead), F types; OHV, OHC, DOHC.

(h) Based on speed : low (<350 rpm), medium (350-1000), high (>1000 rpm)

(i) Based on use : automobile, marine, stationary, aircraft, locomotive.

1.7 Parts of an I.C. Engine

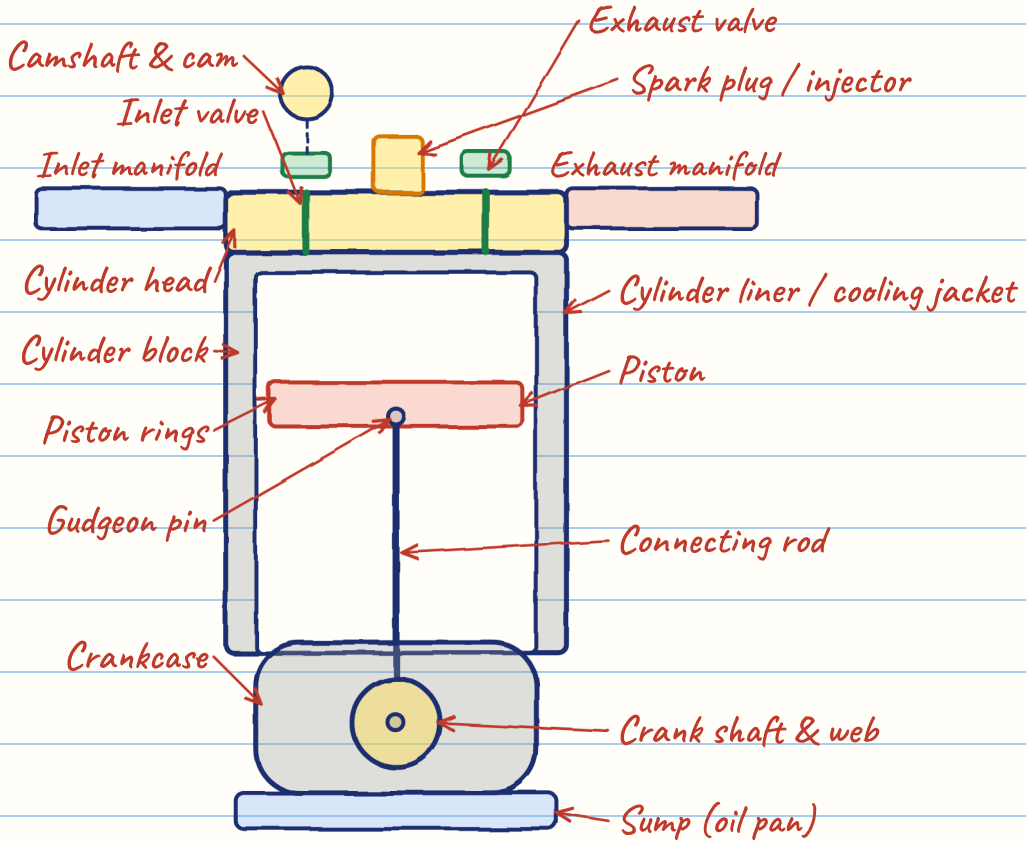


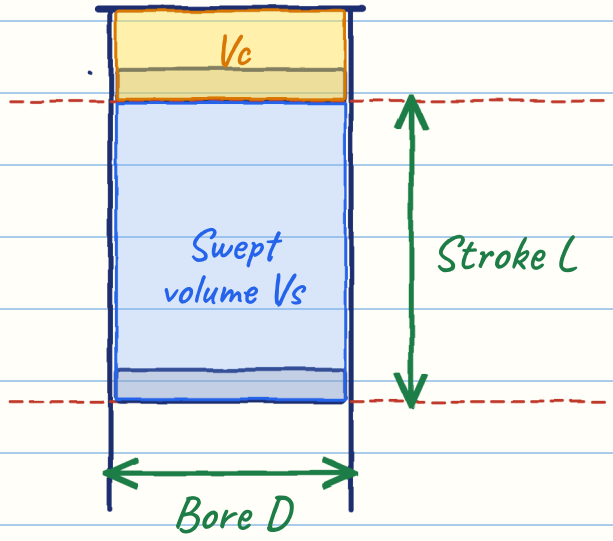
Fig: Main parts of a reciprocating IC engine (schematic)

1.7 Parts - functions (contd.)

- **Cylinder block** : main body; houses the cylinders; made of cast iron / aluminium alloy; has water jackets or cooling fins.
- **Cylinder liner** : replaceable sleeve (dry / wet) inside block, gives wear-resistant surface.

- **Cylinder head** : closes the top end; holds valves, spark plug / injector; combustion chamber formed here. Gasket seals the joint.
- **Piston** : reciprocating part; transmits gas force to the connecting rod; cast Al-alloy, light but strong.
- **Piston rings** : (a) compression rings - seal gas leakage, (b) oil control ring - scrape excess oil from wall.
- **Gudgeon (piston) pin** : joins the piston to small end of connecting rod.
- **Connecting rod** : I-section forged steel; converts reciprocating motion → rotary; big end on crank pin, small end on gudgeon pin.
- **Crankshaft** : converts to rotary motion; carries counterweights; supported by main bearings.
- **Flywheel** : heavy wheel on crankshaft; stores energy in power stroke and releases in the other strokes → smooth rotation.
- **Crankcase** : lower casing enclosing crankshaft. **Sump** holds lubricating oil.
- **Camshaft & cam** : opens inlet / exhaust valves in correct sequence; driven by timing gear / chain / belt at ½ crank speed.
- **Valves** : poppet type; inlet (large head) & exhaust (made of heat-resisting steel); spring-loaded.
- **Manifolds** : inlet manifold supplies mixture/air; exhaust manifold collects burnt gases.
- **Carburettor / Injector** : prepares the air-fuel mixture (SI) / injects fuel (CI).
- **Spark plug** : produces spark in SI engine.

1.8 Engine Terminology



Clearance volume Vc (above TDC piston)
TDC (top dead centre)

$V_s = (\pi/4)\, D^2 L$

$r = (V_s + V_c) / V_c$

$r = 1 + V_s/V_c$

BDC (bottom dead centre)
Total volume $V_1 = V_s + V_c$
Piston speed $S_p = 2LN$ (N in rps)

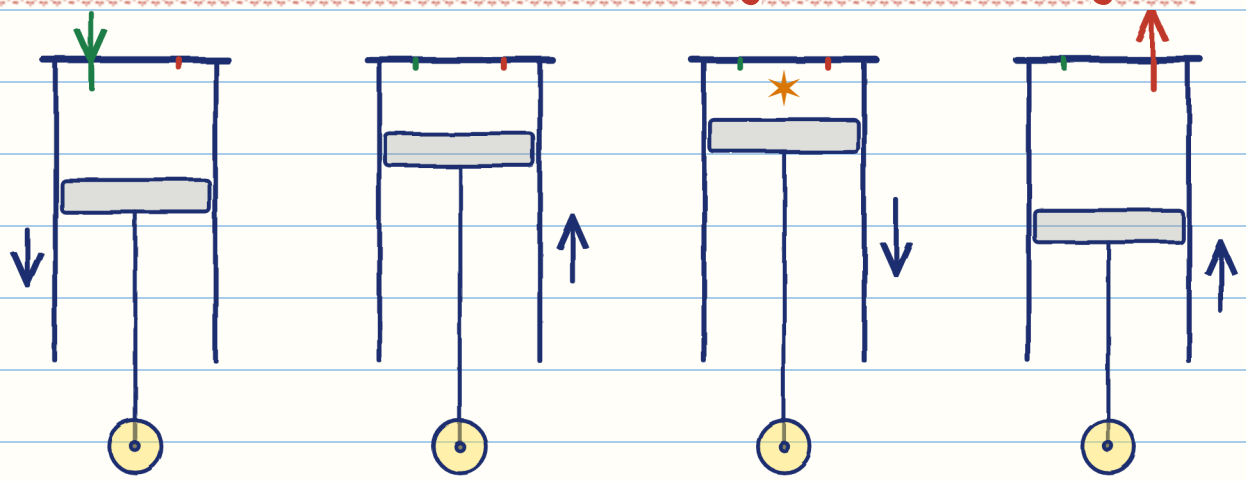
- **TDC** : extreme top position of piston (inner dead centre, IDC).
- **BDC** : extreme bottom position (outer dead centre, ODC).
- **Stroke (L)** : distance travelled by piston between TDC and BDC.
- **Bore (D)** : inside diameter of cylinder.
- **Clearance volume (Vc)** : volume above piston at TDC.
- **Swept volume (Vs)** : volume swept by piston in one stroke.
- **Compression ratio (r)** : ratio of total volume to clearance volume.
  SI : 6-10 ; CI : 15-22.

Displacement of engine = Vs × number of cylinders (cc). Other terms: Mean piston speed = 2LN.

Example

- D = 100 mm, L = 120 mm, Vc = 150 cc. Find r.
  → $V_s = (\pi/4)(10)^2(12) = 942.5$ cc ; $r = 1 + 942.5/150 = 7.28$

1.9 Four-Stroke SI (Petrol) Engine - Otto Cycle



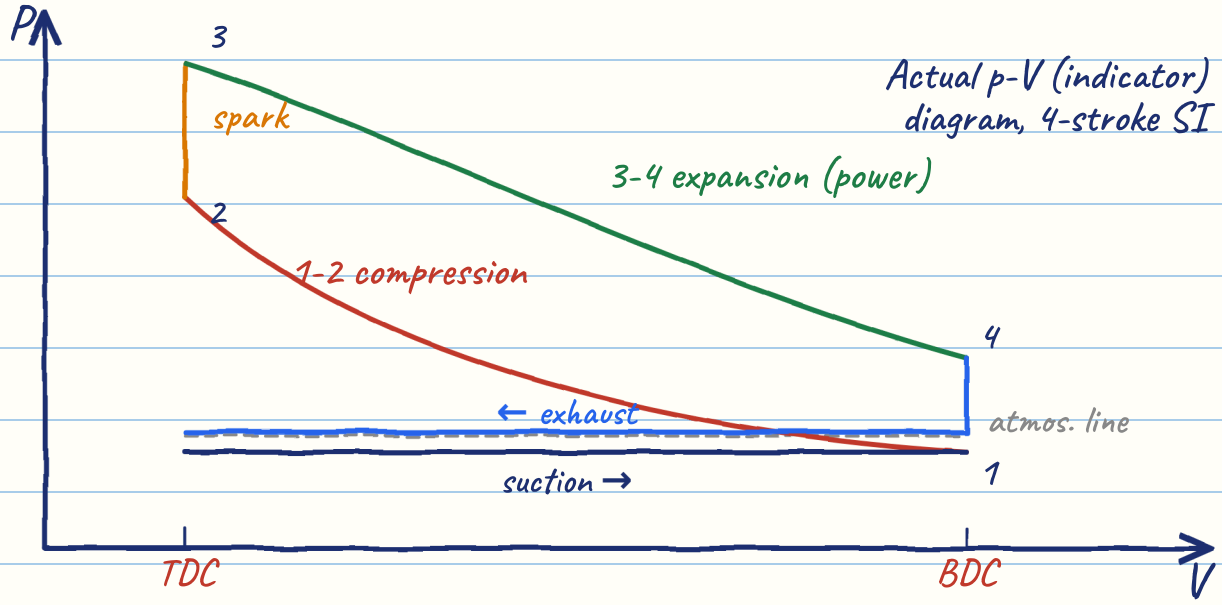
1. SUCTION 2. COMPRESSION 3. POWER 4. EXHAUST

Both valves shut in compression & power; piston: ↓ ↑ ↓ ↑ ; crank turns 720° per cycle

Working (4 strokes = 2 revolutions)

- **1. Suction stroke** : inlet valve open, exhaust closed; piston goes TDC → BDC; air-fuel mixture (from carburettor) drawn in at about 0.8-0.9 atm.
- **2. Compression stroke** : both valves closed; piston BDC → TDC; mixture compressed to ≈ 8-10 bar; near the end (≈ 20-30° before TDC) the spark plug fires.
- **3. Power / expansion stroke** : combustion raises pressure to 35-45 bar, T ≈ 2500 K; gases push piston TDC → BDC; **only stroke producing work**. Exhaust valve opens near the end (EVO).
- **4. Exhaust stroke** : exhaust valve open; piston BDC → TDC pushes burnt gases out at ≈ 1.1-1.2 atm.

1.9 (contd.) p-V diagram of 4-stroke SI engine



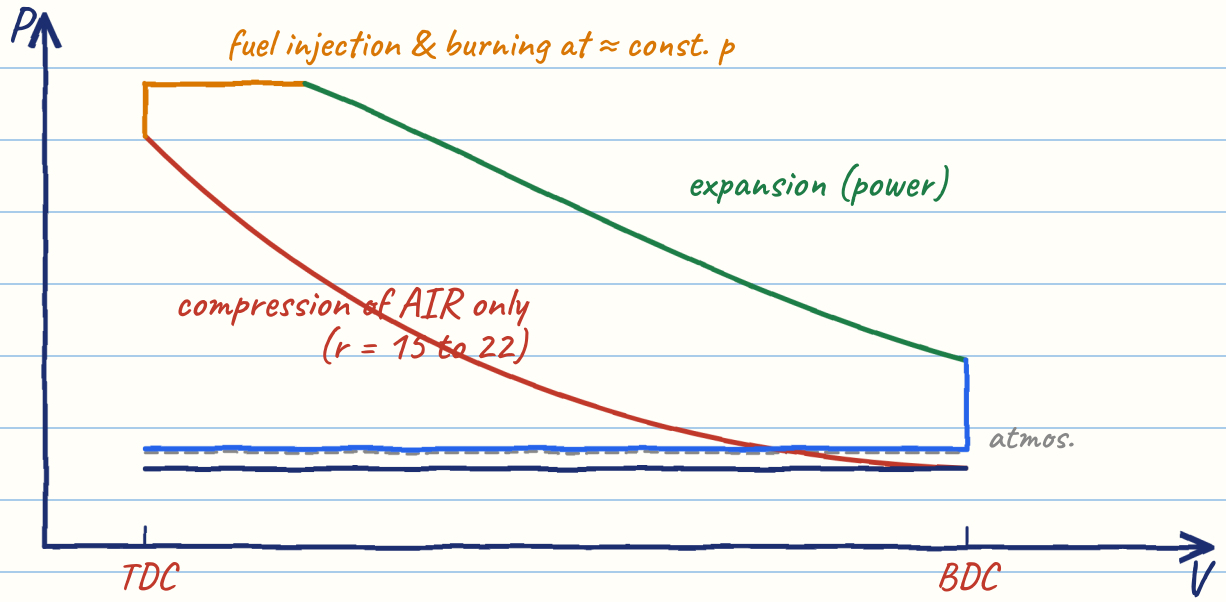
- Area of upper loop = **work done** (positive); lower loop (suction & exhaust) = pumping work (negative), small.
- Net work = Area(upper) − Area(lower).
- **Heat is added at nearly CONSTANT VOLUME** (2→3) because combustion is rapid.
- Only 1 power stroke in every 2 revolutions ⇒ power strokes per minute = N/2.

Valve operation in 4-stroke engines

- Each valve opens once in 2 revs, so camshaft turns at **half** the crank speed.
- Cam, follower, push rod, rocker arm, valve spring - valve train.
- Gear ratio crank : cam = 2 : 1.

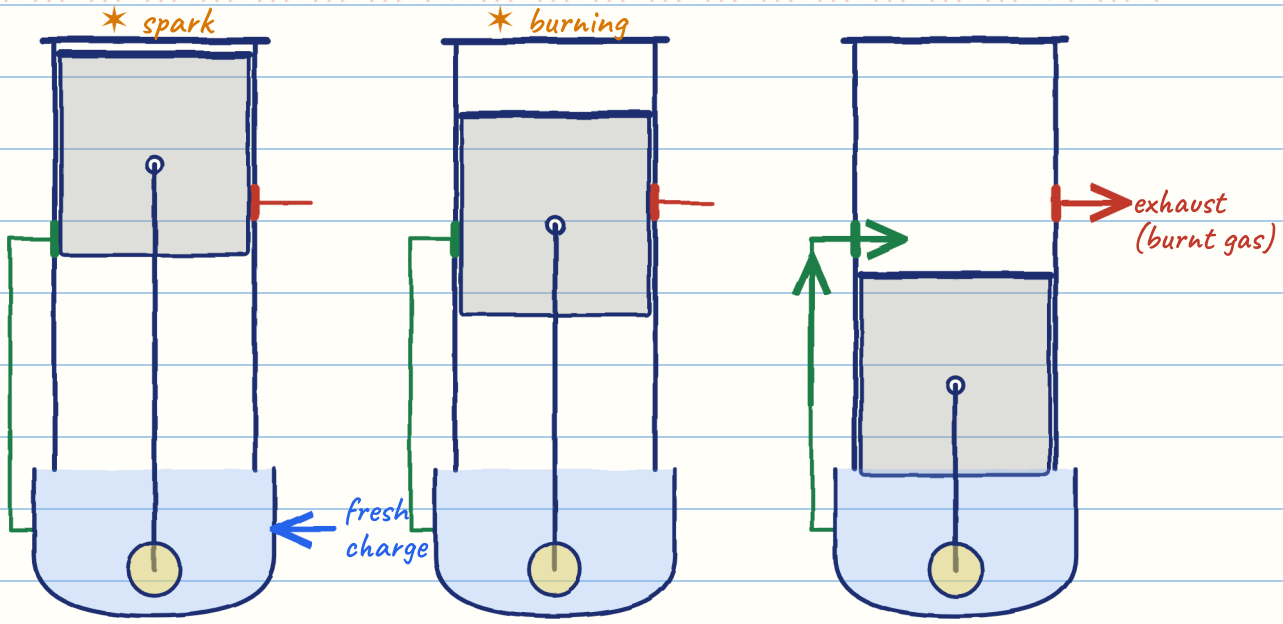
1.10 Four-Stroke CI (Diesel) Engine

- Intake of **air only** (no throttle) → compressed to 30-40 bar, T = 600-800 °C; fuel injected near end of compression; self-ignites ⇒ no spark plug.



- **Suction** : only air enters (inlet open). **Compression** : high compression ratio 15-22 raises temperature above self-ignition temp. of diesel.
- **Power** : fuel (≈ 10-15° bTDC) injected at 150-250 bar; burns at roughly const. pressure; expansion pushes piston.
- **Exhaust** : burnt gases pushed out.
- Load is controlled by *quantity of fuel* (quantity governing), not by throttling air.

1.11 Two-Stroke Engine (crankcase scavenged)



(a) Upstroke: compression + crankcase suction

(b) Downstroke: power + crankcase compression

(c) Exhaust & transfer (scavenging)

- Cycle in **one revolution** : power stroke at every revolution → 2 strokes (up & down). No valves - ports (transfer, exhaust, and inlet - piston-port or reed valve) are covered / uncovered by the piston.
- **Upward stroke (a)** : piston covers transfer & exhaust ports → compresses the charge above; at the same time creates vacuum in crankcase → fresh charge enters through inlet port. Spark near TDC.
- **Downward stroke (b)** : combustion → power stroke; piston covers inlet port & compresses charge in crankcase (≈ 1.3-1.5 bar).
- (c) Exhaust port opens first (blow-down), then transfer port opens → pressurised charge from crankcase flows to cylinder, deflector on piston pushes burnt gas out (**scavenging**).

1.12 Scavenging in 2-stroke engines

- Scavenging = process of sweeping out burnt gases with fresh charge.
- Types : (1) **Cross-flow** - deflector piston; (2) **Loop** - ports on same side, charge loops (Schnuerle / Curtis); (3) **Uniflow** - charge enters at one end & exhaust at other end through valves/ports (best, used in large diesels).
- Problem in petrol 2-stroke : part of fresh charge escapes into the exhaust ⇒ fuel loss, HC emissions, so fuel injection is used now.

Comparison : 4-stroke vs 2-stroke

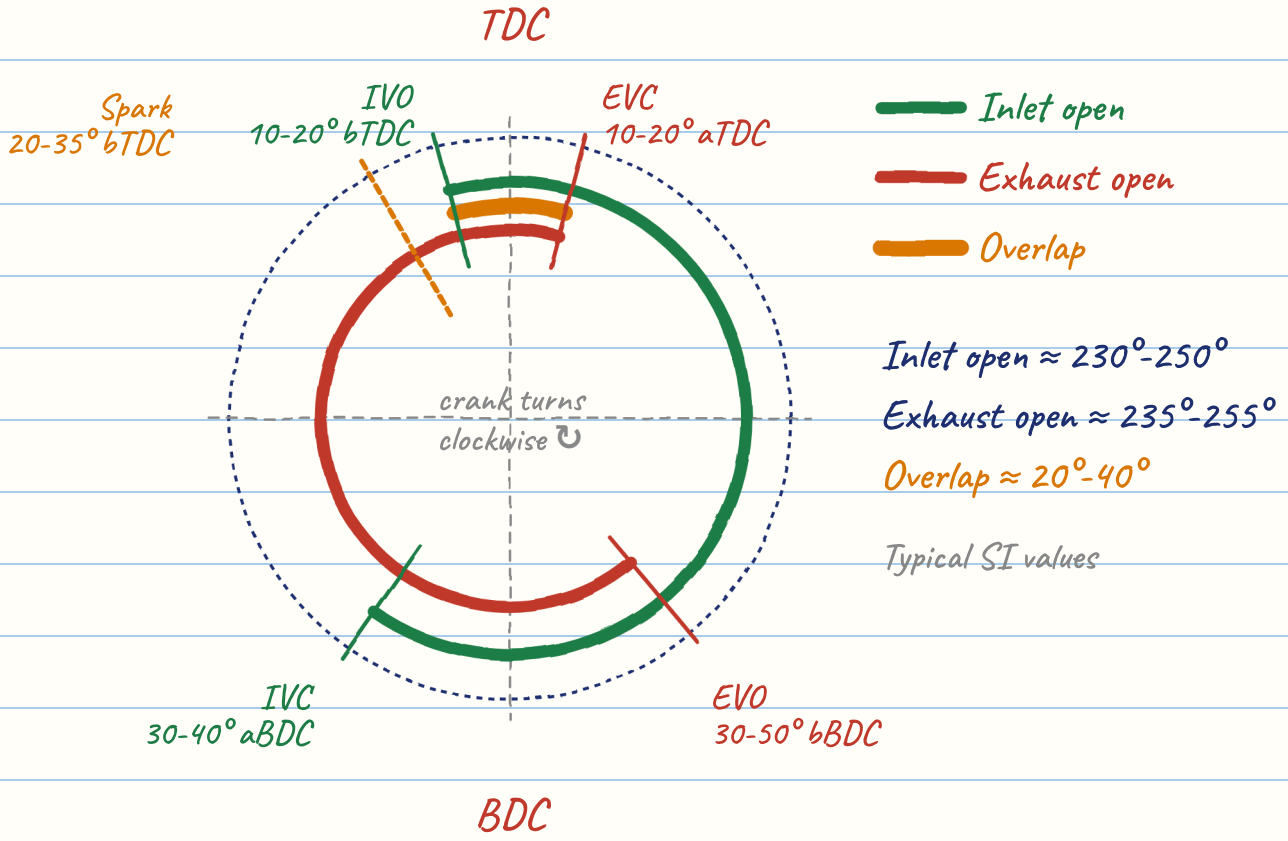
Point	4-stroke	2-stroke
Cycle	4 strokes / 2 revs	2 strokes / 1 rev
Power stroke	once in 2 revs	once every rev
Power for same size	less	≈ 1.6-1.8 × more (not 2×)
Valves	poppet valves + cam	ports only (simple)
Flywheel	heavy	light
Thermal efficiency	higher	lower (fuel loss)
Fuel consumption	lower	higher
Lubrication	sump / pressure	petrol-oil mix
Cooling	easier	harder (more heat)
Wear	less	more
Used in	cars, trucks, buses	scooters, mopeds, outboards

1.13 SI Engine vs CI Engine

Point	SI engine	CI engine
Cycle	Otto (const. volume)	Diesel / Dual
Fuel	petrol (volatile, high self-ignition temp.)	diesel (less volatile, low SIT)

Point	SI engine	CI engine
Intake	air + fuel mixture	air only
Fuel supply	carburettor / MPFI	fuel injection pump & injector
Ignition	spark plug (electrical)	self ignition by hot air
Compression ratio	6 - 10	15 - 22
Max. pressure	35 - 45 bar	60 - 100 bar
Weight	lighter	heavier, stronger
Efficiency	25 - 30 %	35 - 40 %
Speed	high (up to 6000+ rpm)	lower (up to 4000)
Load control	quantity of mixture (throttle)	quantity of fuel
Running cost	higher	lower
Noise / vibration	less	more
Emission	CO, HC	NOx, smoke, PM
Starting	easy	harder in cold

1.14 Valve Timing Diagram (4-stroke)



- It shows the **angular position of crank** (w.r.t. TDC & BDC) at which valves open and close.

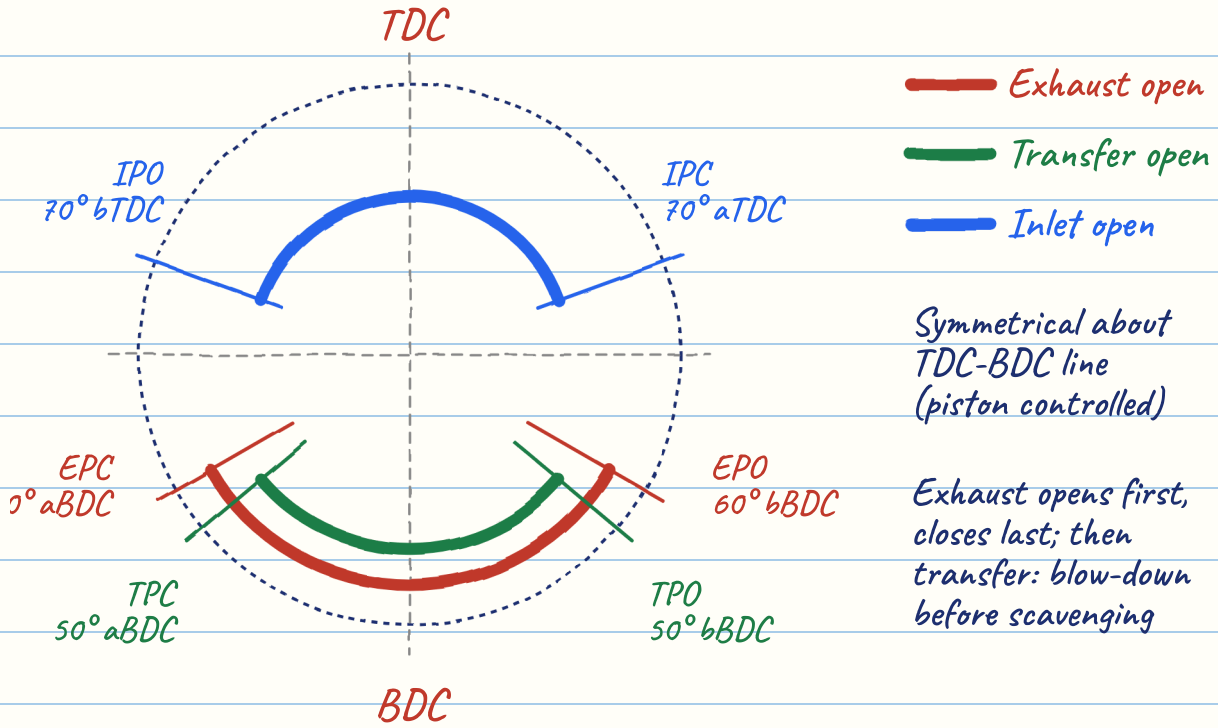
1.14 Valve timing (contd.)

- **Actually**, inlet valve opens **before TDC** (IVO, 10-20°) - **lead** - so that it is fully open when the suction stroke starts, giving full charge.
- Inlet valve closes **after BDC** (IVC, 30-40°) - **lag** - since the incoming charge still has momentum (ram effect), more charge enters even though the piston is rising.
- Exhaust valve opens **before BDC** (EVO, 30-50°) - **lead** - so that pressure falls (blow-down) and the piston has less back pressure in exhaust stroke.
- Exhaust valve closes **after TDC** (EVC, 10-20°) - **lag** - to get complete scavenging by inertia of exhaust gas.

Valve overlap = (IVO lead + EVC lag) ≈ 20-40° when both valves are open near TDC : good scavenging, better volumetric efficiency; but in petrol engine too large overlap wastes fuel.

- **Effect of speed** : at higher speeds greater lead and lag are given because time available is less.
- Diesel values : IVO 10-25° bTDC, IVC 25-40° aBDC, EVO 35-45° bBDC, EVC 10-20° aTDC, injection 10-15° bTDC.
- Diesel engine overlap is larger (pure air escapes, no fuel loss); also helps cooling of valves/piston crown.
- Ignition (SI) 20-35° before TDC and fuel injection (CI) 10-15° before TDC are also marked on the diagram.

I.15 Port Timing Diagram (2-stroke)



- Shows the crank angles at which the **ports** open & close for piston-controlled 2-stroke engine.
- **Exhaust port** opens earlier (EPO ≈ 60° bBDC) and closes later (EPC ≈ 60° aBDC) than the transfer port (TPO/TPC ≈ 50°).

- *Inlet port* opens ≈ 70° bTDC and closes ≈ 70° aTDC (as piston covers it).
- Diagram is symmetrical because the piston itself uncovers & covers ports (fixed timing) - for unsymmetrical timing, reed / rotary valves are used.

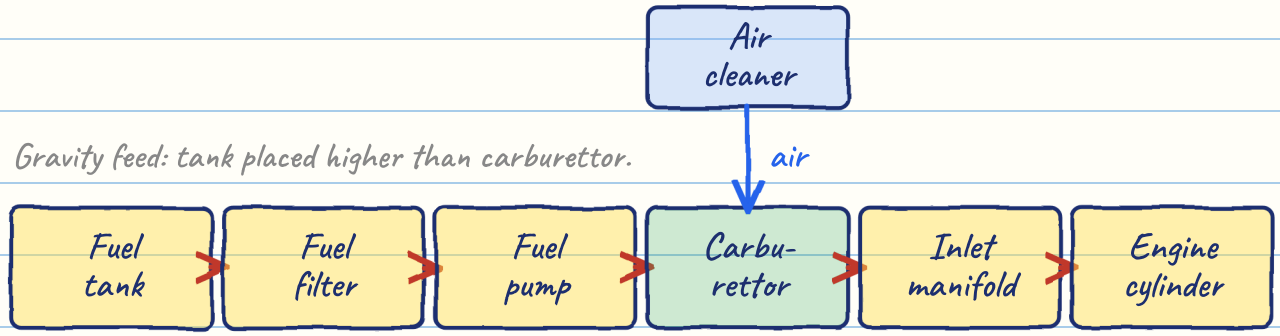
Quick Revision

- $r = 1 + Vs/Vc$; $Vs = (\pi/4)D^2L$; 4-S: 720° ; 2-S: 360°
- SI → Otto, spark, r = 6-10 ; CI → Diesel, auto-ignition, r = 15-22

UNIT - II : FUEL, COOLING, LUBRICATION & IGNITION SYSTEMS

2.1 Fuel Supply System of an S.I. Engine

- Function : store the fuel, deliver it at the correct rate and prepare a combustible air-fuel mixture of correct strength for every speed & load.



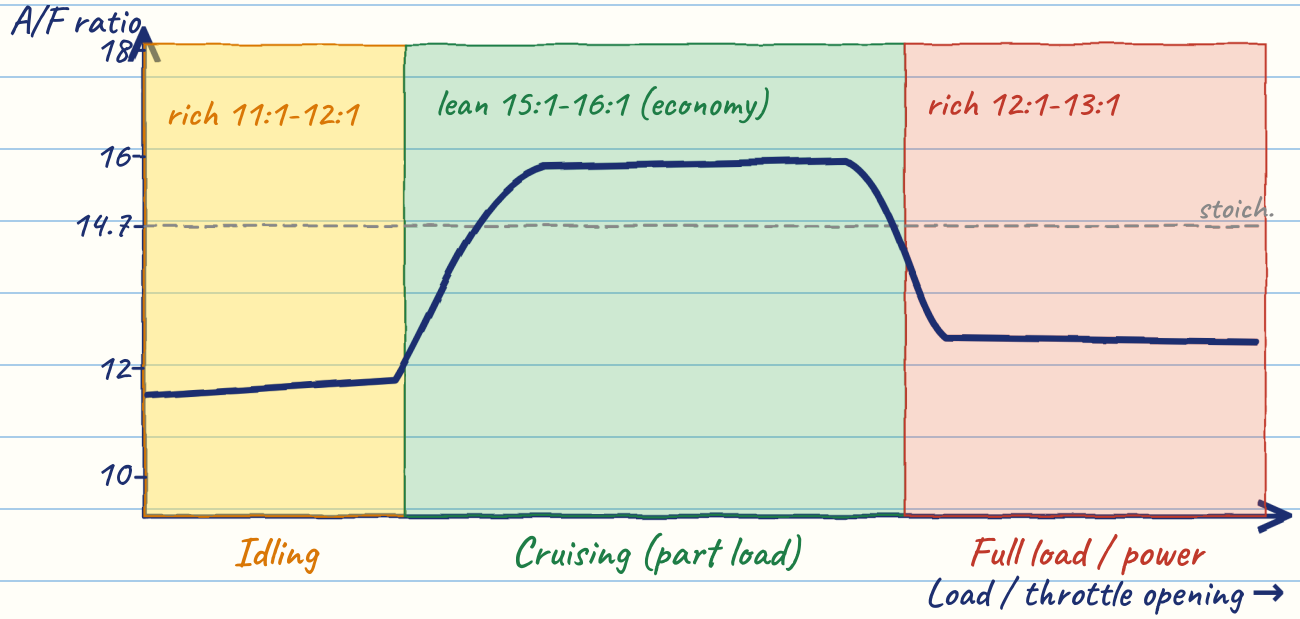
Fuel (petrol) is lifted from the tank by the pump, cleaned in the filter, metered & mixed with filtered air in the carburettor, and the mixture is drawn through the manifold into the cylinder.

- Fuel tank : sheet-steel / plastic, has filler cap with vent, baffles (stop surging) & a float type level gauge.
- Fuel lines : copper / nylon pipes carry fuel from tank → pump → carburettor.
- Strainer / filter : removes dirt, water & gum (paper, ceramic or gauze element).
- Fuel pump : mechanical (diaphragm) or electrical; needed where tank is below the carburettor.
- Carburettor : meters the fuel & atomises it in the air stream.

2.2 Air-Fuel Ratio Requirements

- Complete combustion needs about 14.7 : 1 (stoichiometric, chemically correct) - air : fuel by mass.

- **Idling** - rich (11:1 to 12:1) because residual exhaust gas dilutes the charge.
- **Cruising / part load** - lean (15:1 to 16:1) for economy.
- **Full load / acceleration** - rich (12:1 to 13:1) for maximum power.



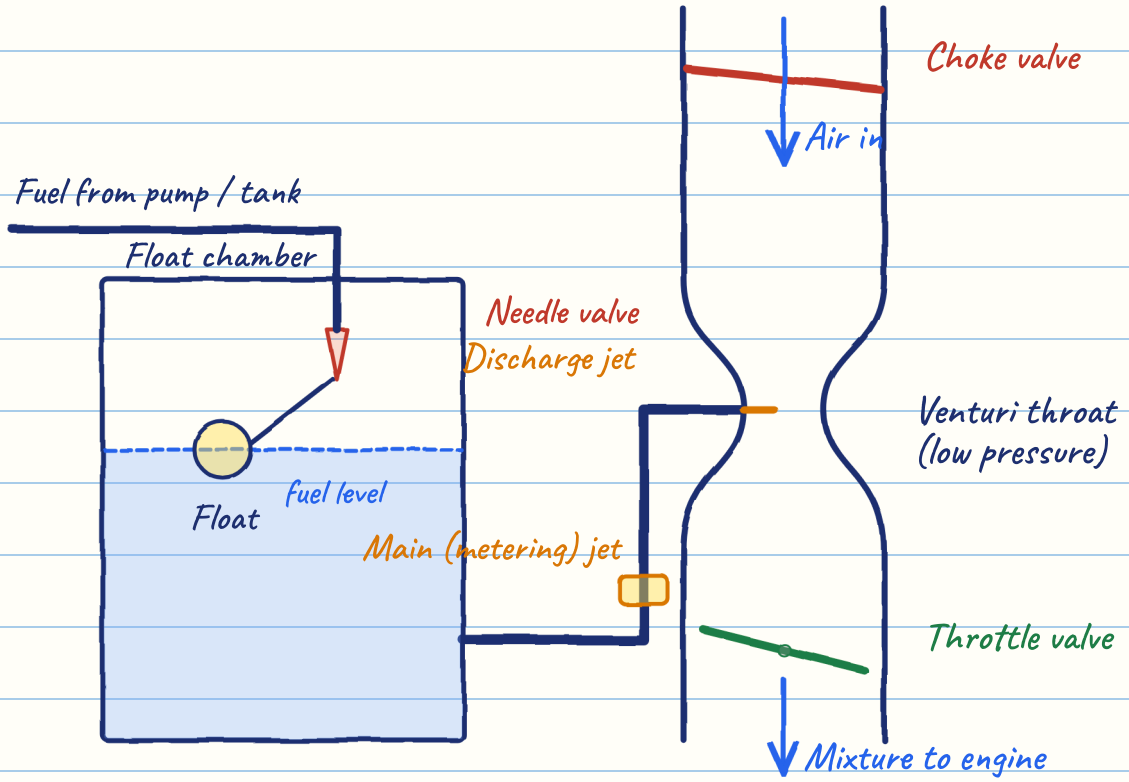
Rich mixture → max. power, more CO & HC. Lean mixture → best economy, higher NOx & cylinder temperature (risk of backfire).

2.3 Carburetion - Principle

- **Carburation** = process of preparing a combustible mixture of petrol & air in the proper ratio outside the cylinder.
- Based on the venturi principle (Bernoulli) : at the throat velocity is high ⇒ pressure falls (partial vacuum) ⇒ fuel is pushed from float chamber through the jet by the atmospheric pressure acting on the fuel surface.

$$\Delta p = \rho_a (C^2/2), \text{ fuel flow} \propto \sqrt{(\Delta p)}$$

2.4 Simple Carburettor



- **Working** : on the suction stroke air is drawn through the venturi; pressure at throat < pressure in float chamber ⇒ fuel is sprayed through the jet & vaporises. Throttle valve controls the quantity of mixture delivered to the engine ⇒ power.
- **Float chamber** keeps fuel level constant (≈ 4-5 mm below jet tip) to prevent fuel spilling when engine is stopped. The needle valve shuts when the float rises.
- **Choke valve** is closed for cold starting → creates strong vacuum → very rich mixture.

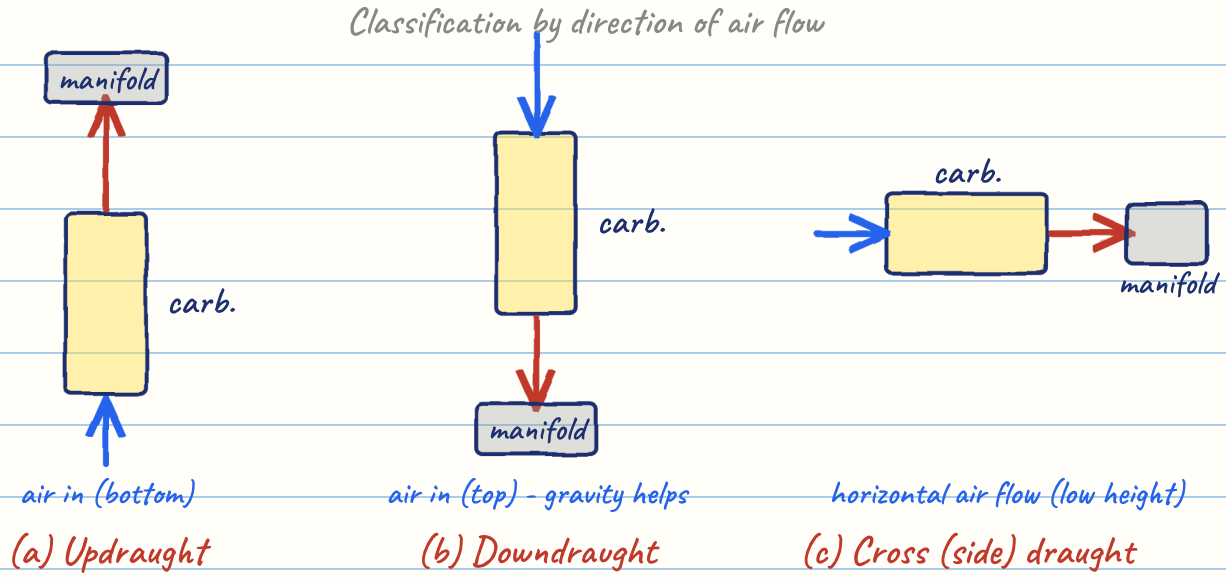
Defects of a simple carburettor

- Gives progressively **richer** mixture as throttle opens (fuel flow ∝ $\sqrt{\Delta p}$ but air flow density falls).
- Cannot give correct mixture for idling, cold start or acceleration, no economy at cruise.

2.5 Additional Devices in a Modern Carburettor

- **Compensating jet** - supplies air-bled fuel at high speeds to keep mixture from becoming rich.
- **Idling (slow-running) jet** - supplies rich mixture past the nearly closed throttle at low speed.
- **Accelerating pump** - a piston/diaphragm pump linked to the throttle which injects an extra shot of petrol for sudden acceleration.
- **Economiser** enriches at full load; **auto-choke** (bimetal) for cold start; **air bleed** improves atomisation.

2.6 Types of Carburettor

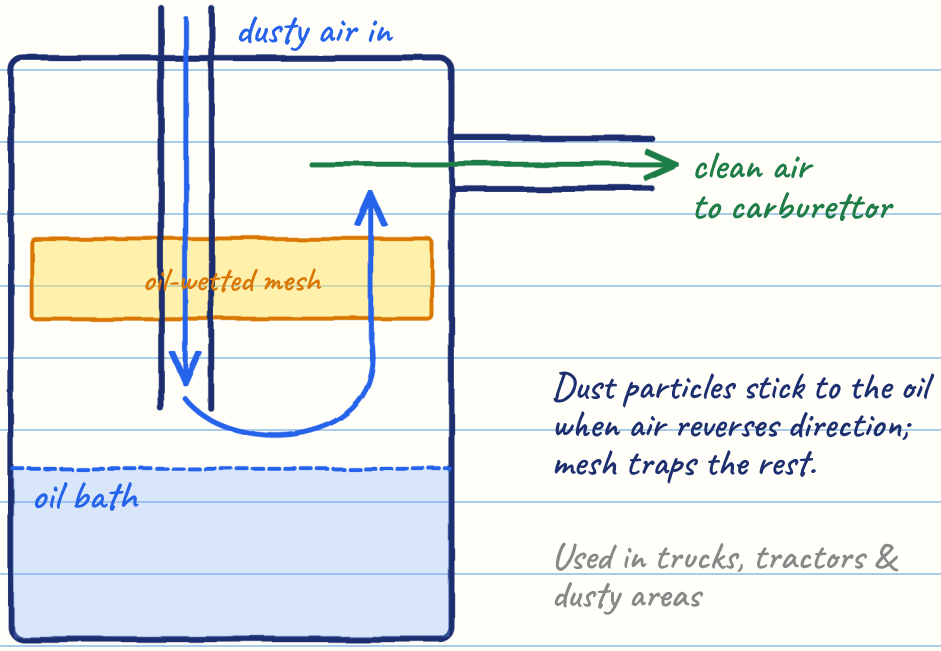


- (1) **By direction of airflow** : updraught, downdraught (most common - gravity helps fuel flow, easier cold start, better volumetric efficiency) and cross-draught.
- (2) **By venturi type** : **constant choke** (Solex, Zenith, Carter - area constant, depression varies) and **constant vacuum** (S.U., Constant Depression type - a piston moves to keep vacuum constant, variable choke).

- (3) Single, dual & multi-barrel types. **Solex** : downdraught constant-choke ; **S.U.** : horizontal variable-choke with tapered needle.

2.7 Air Filters (Air cleaners)

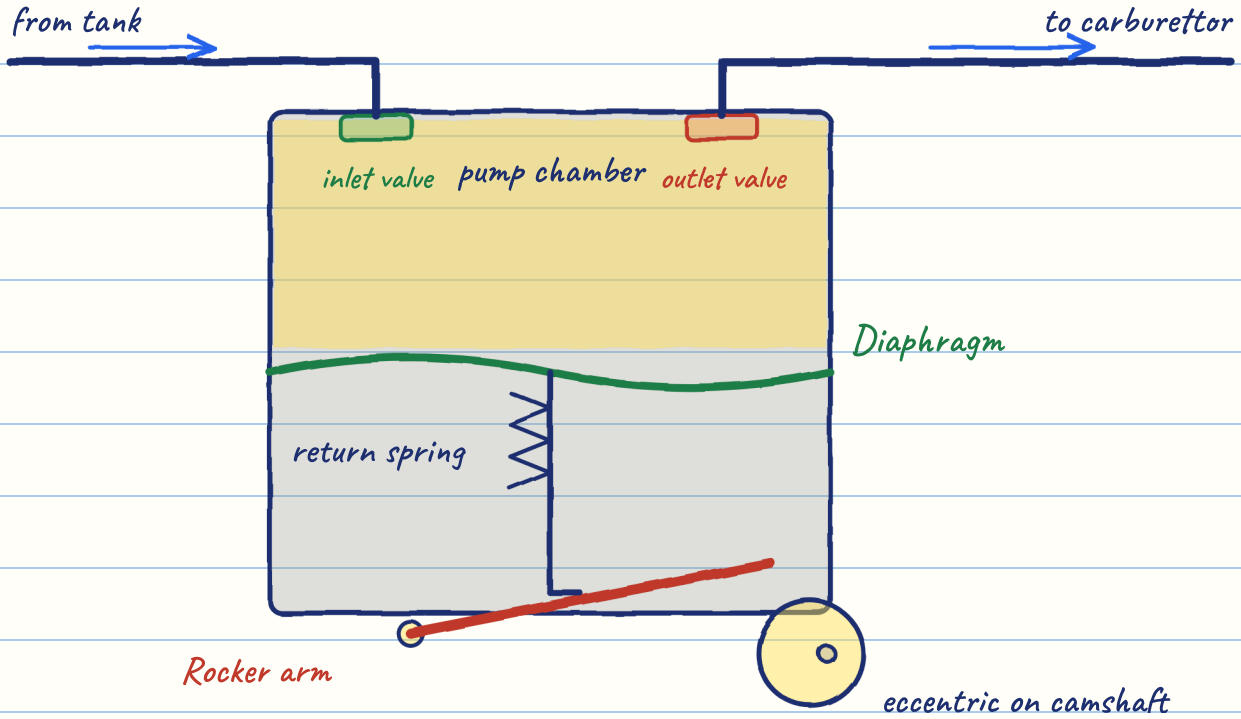
- Purpose : remove dust from the intake air (dust causes cylinder / ring wear) and silence intake noise.
- **Dry (paper) type** : pleated resin-impregnated paper element; cheap, efficient (99%), replaced periodically; most cars.
- **Oil-bath type** : air strikes oil surface, dust sticks to oil; wire mesh wetted with oil traps the rest; used in heavy duty / dusty places.
- **Viscous / centrifugal** types : oil-wetted mesh, cyclone pre-cleaners for trucks.



2.8 Fuel Pumps

- Needed when tank is below the carburettor (or fuel is injected under pressure).

(a) Mechanical (diaphragm) pump



- Driven by an eccentric on the camshaft through a rocker arm.
- **Suction stroke** : arm pulls diaphragm down against spring; vacuum opens inlet valve, fuel enters chamber from tank.
- **Delivery stroke** : when cam releases the arm the spring pushes the diaphragm up → inlet valve closes, outlet valve opens → fuel flows to carburettor. If the carburettor float needle is closed, the diaphragm stays down - pump idles.
- Delivery pressure ≈ 0.2 - 0.3 bar (low pressure). Simple, cheap, but vapour-lock problems.

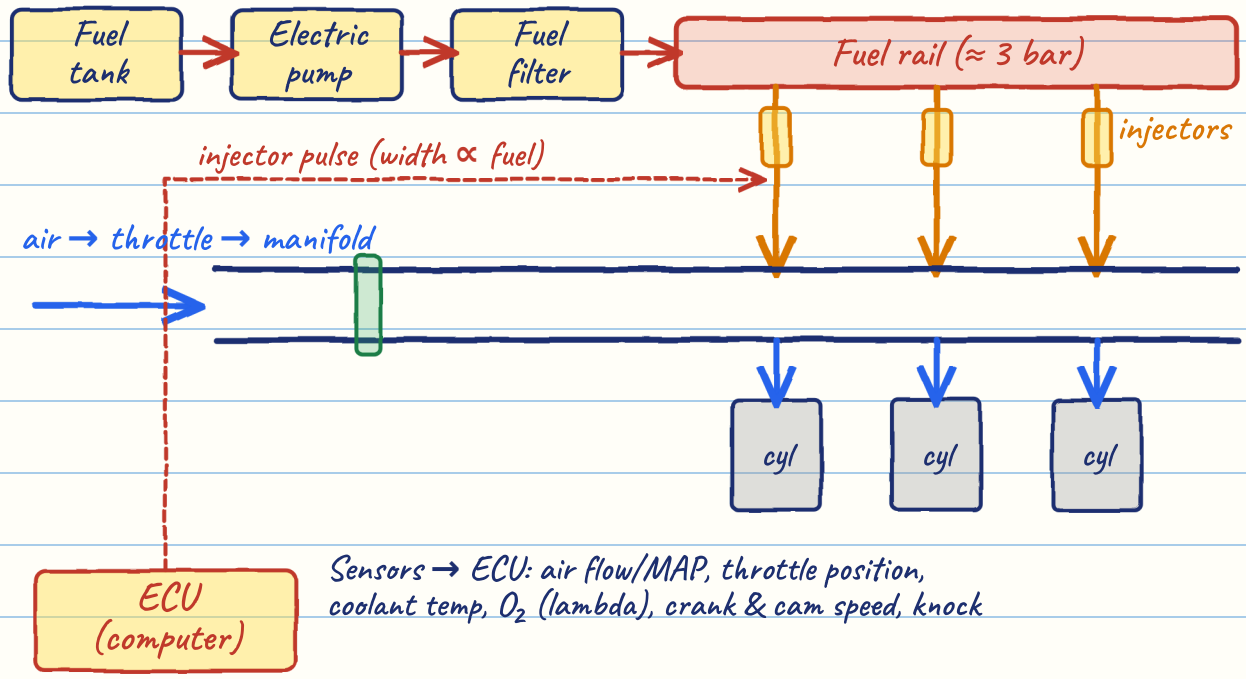
(b) Electrical fuel pump

- A **solenoid** pulls an armature/diaphragm; contact points open & close automatically (like a buzzer) giving reciprocating action. Fitted near the tank (pushes fuel ⇒ no vapour lock). In fuel-injected cars an **in-tank rotary (roller-cell / gerotor) pump** gives 3-5 bar.

- Advantages : delivers fuel even before the engine starts; can be placed anywhere; less vapour lock. Disadvantage : needs battery, noise, points burn.
- Fuel filter (2.9) : paper element (5-10 μm) / gauze between tank-pump-carburettor, replaced every 20-30 thousand km.

2.10 Gasoline (Petrol) Injection Systems

- Fuel is injected under pressure; computer meters it exactly. Advantages over carburettors : better mixture distribution, higher volumetric efficiency (no venturi restriction), precise A/F control for emission norms, better cold start & acceleration, ~ 10 % less consumption.



- Types : (a) Throttle body (single point) injection - TBI : 1 injector above the throttle. (b) Multi-point (port) injection - MPFI : injector for each cylinder at the inlet port; sequential or group firing. (c) Gasoline Direct Injection - GDI : injector sprays straight into the cylinder at 50-200 bar; stratified lean charge, ~ 15-20 % better economy.

- ECU reads sensors (MAF/MAP, TPS, coolant temp, lambda / O_2, crank speed) & varies injector pulse width. Closed loop with the O_2 sensor keeps $\lambda \approx 1$ for the 3-way catalytic converter.

2.11 Cooling System - Requirements

- About 30 - 35 % of the heat of fuel goes to the coolant & walls, 30-35 % to exhaust, 30 % to useful work.
- Gas temperature reaches 2000-2500 °C but metal & lubricating oil limits are low ⇒ cooling is must to : (1) avoid overheating & seizure of piston, (2) prevent oil film breakdown, (3) avoid pre-ignition / knocking, (4) keep valve & head strength, (5) maintain high volumetric efficiency.
- But over-cooling wastes heat, lowers thermal efficiency, causes fuel condensation & wear. Optimum cylinder wall temp. ≈ 80-95 °C (water outlet).

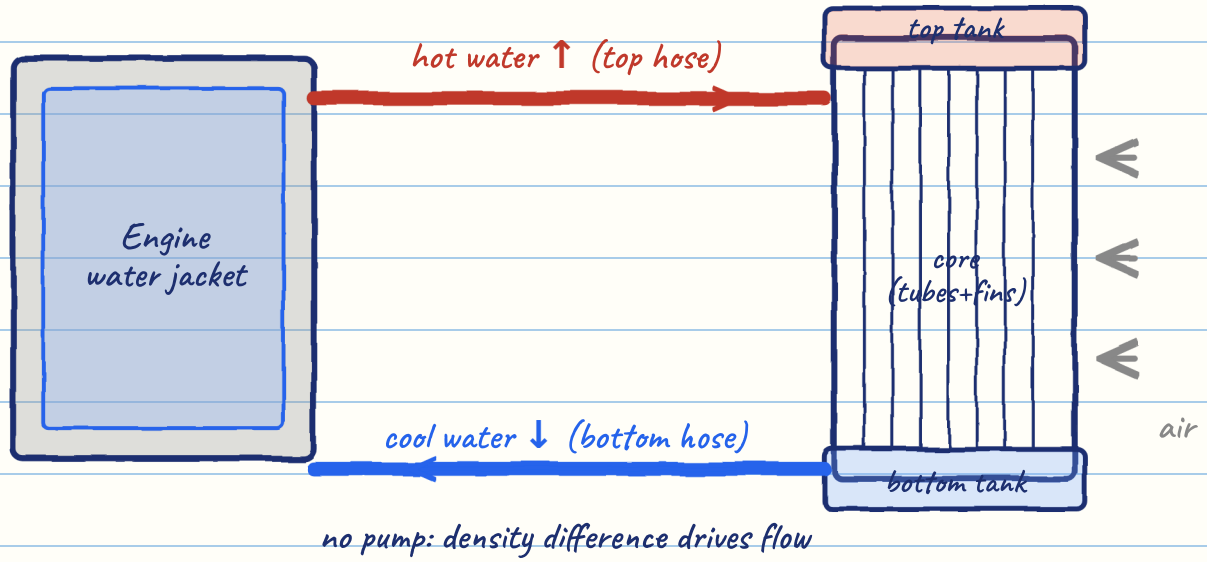
2.12 Air Cooling

- Heat is removed by air flowing over fins cast on the cylinder & head. Used in motorcycles, scooters, small engines & aircraft.
- Fins : greater surface, thin & closely spaced; fan / blower and cowling (baffles) are used in engines.
- Advantages : light, simple, no radiator / coolant leakage / freezing, quick warm-up. Disadvantages : non-uniform cooling, noisy, lower power density, not suitable for big engines.

2.13 Liquid (Water) Cooling

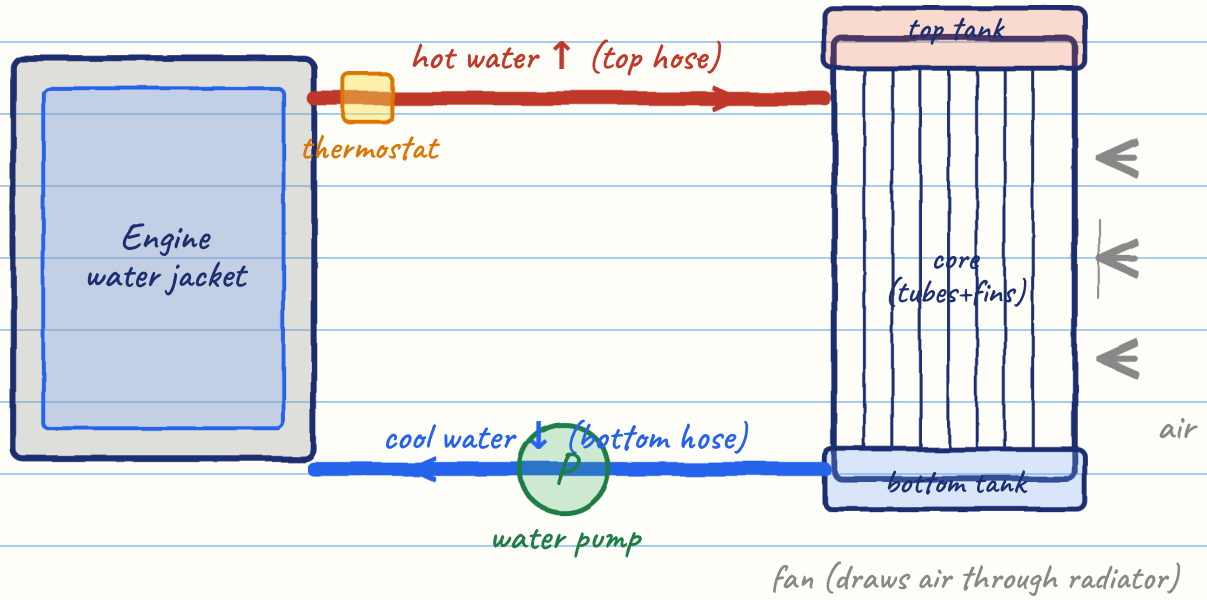
- Water jackets surround cylinders & head; heat taken by water → radiator → air. Coolant = water + ethylene glycol (anti-freeze) + corrosion inhibitor.

(a) Thermo-siphon system



- Water in the jacket heats, becomes lighter & **rises** to radiator top tank; cooled water is denser & **sinks** to bottom tank → back to engine. Flow due to density difference only.
- **No pump**, simple, but slow circulation, needs large water capacity & high radiator position ⇒ obsolete; used in old tractors.

(b) Forced circulation (pump) system



- **Water pump** (centrifugal, belt driven) circulates water at 20-60 l/min.
- **Thermostat** (wax or bellows type) stays closed until ≈ 82 °C - gives quick warm-up, then opens to the radiator; a bypass keeps flow in the block when closed.
- **Radiator** : top & bottom tanks joined by tubes with fins (cellular / tubular core), pressure cap (0.9-1 bar) raises boiling point. **Fan** (belt or electric) pulls air.
- Liquid cooling : uniform, quiet, higher power density (cars, buses) vs air cooling : light & simple (bikes).

2.14 Lubrication System

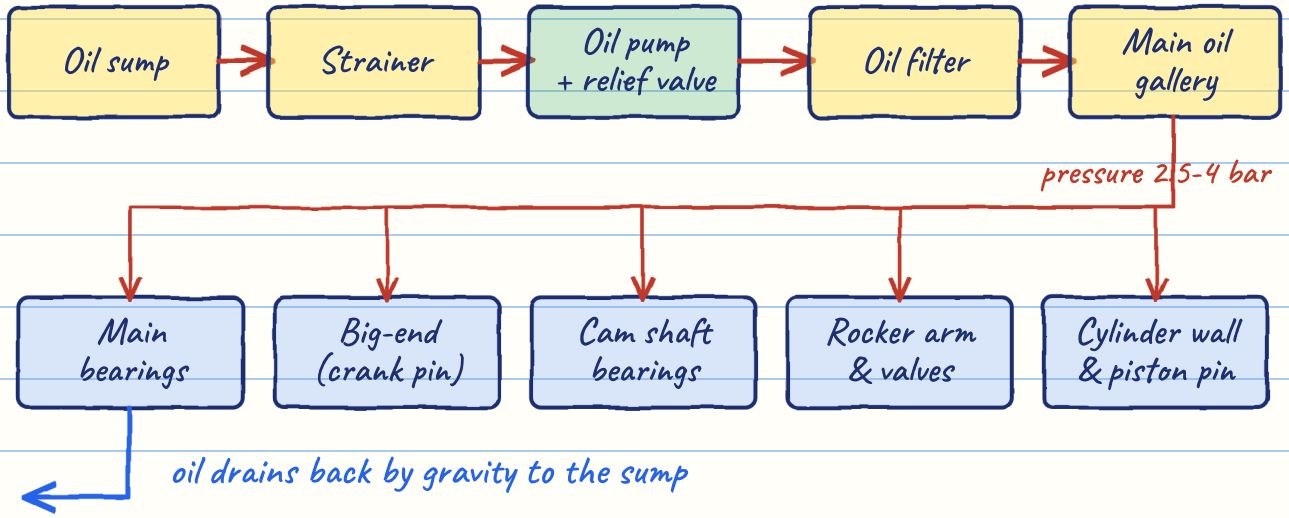
- **Functions of lubricant** : (1) reduce friction & wear, (2) cooling of piston & bearings, (3) cleaning (carries away wear particles), (4) sealing of piston rings, (5) cushioning shocks, (6) corrosion protection.
- Properties of lube oil : suitable **viscosity** & high viscosity index (VI), flash point, pour point, oxidation stability, detergent / dispersant additives, low volatility. SAE grades 10W-30, 20W-40.

Types of lubrication systems

(1) Splash / "Flash" system

- Connecting-rod **dippers** scoop oil from troughs in the sump / crankcase at each revolution & splash it on cylinder walls, piston pin, main & cam bearings. Used in small, low speed engines, lawn mowers. Oil level must be kept constant; variation in splash with speed.

(2) Pressurised (forced feed) system



- **Oil pump** (gear / rotor type) draws oil from sump through the strainer & forces it at 2.5-4 bar through the filter to the main gallery & drilled passages to bearings, cams, rockers; piston & cylinder wall are lubricated by oil thrown from the big-end. **Relief valve** limits the pressure.
- **Wet sump** : oil stored in the sump below the engine (cars). **Dry sump** : oil collected in a separate tank by a scavenge pump (racing, aircraft; better when tilted).
- Combination of splash + pressure also used (semi-pressure).

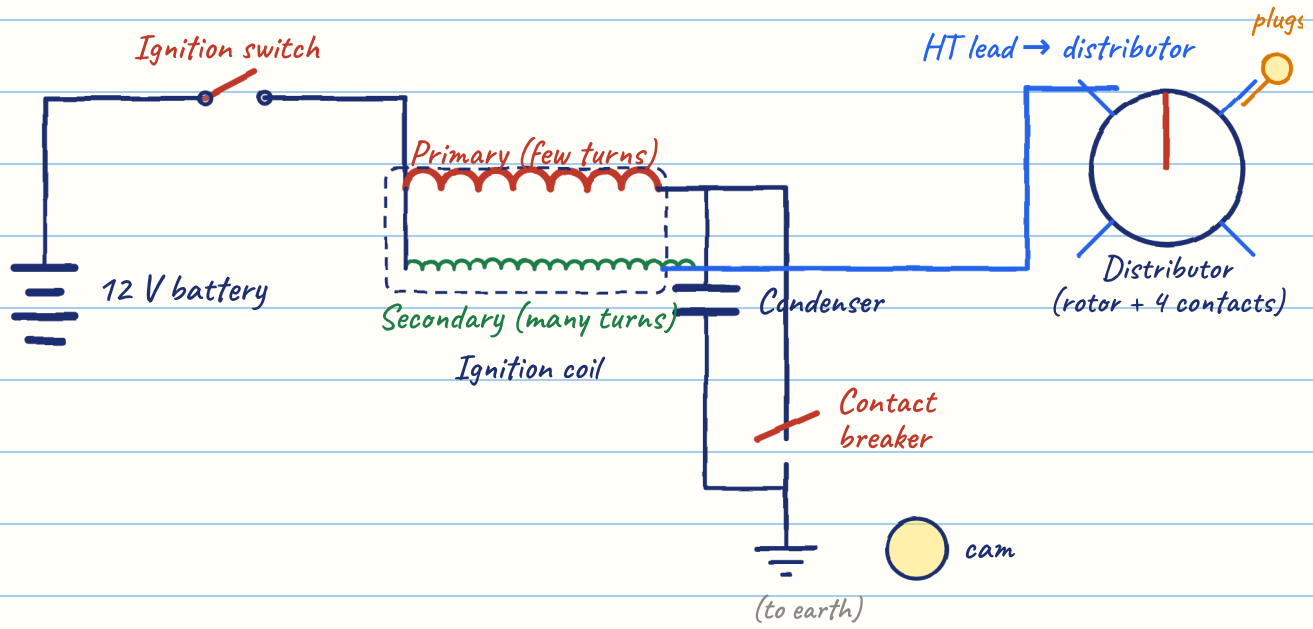
(3) Mist (petroil) lubrication

- Used in **two-stroke** engines : lube oil (2-6 %) is mixed with the petrol (or injected into intake). The charge mist passes through the crankcase; oil droplets deposit on parts, remaining oil is burnt with the fuel. Simple, light, **no sump**; but gives smoke & carbon deposits, poor control.

2.15 Ignition System

- **Function** : to supply a high-voltage spark (≈ 8,000 - 30,000 V) at the right instant (at the end of compression) to every cylinder in the **firing order** (4-cyl : 1-3-4-2 or 1-2-4-3).
- **Requirements** : (1) adequate spark energy & voltage, (2) correct timing with advance as per speed & load, (3) distribute in firing order, (4) reliable, durable, easy to maintain.

2.16 Battery (Coil) Ignition System



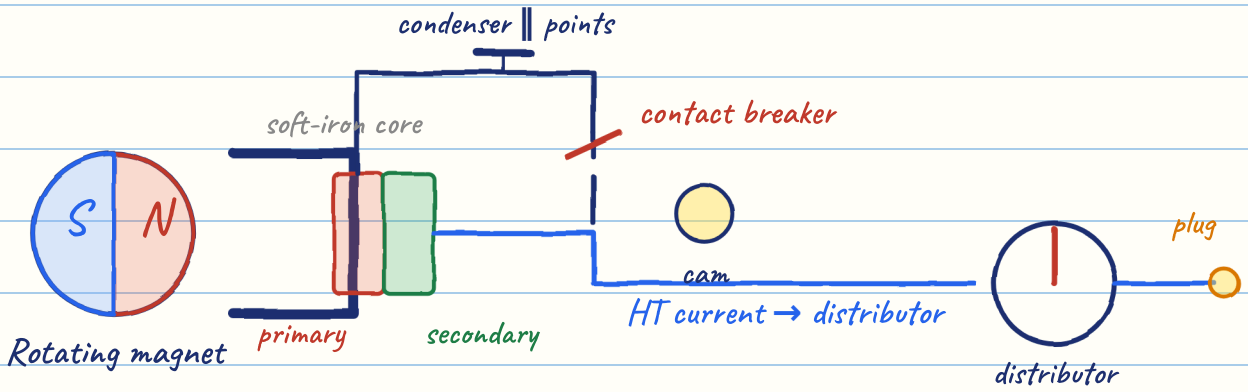
- Components : battery, ignition switch, **ignition coil** (auto-transformer), **contact breaker** & cam, **condenser**, **distributor**, spark plugs.
- **Working** : with points closed, current ≈ 3-4 A flows in the primary building a magnetic field. The cam opens the points → primary current **collapses suddenly** → flux collapses & cuts the many secondary turns ⇒ high voltage (≈ 20-30 kV) induced in the secondary → rotor → distributor contact → spark plug → spark jumps the gap (0.6-0.8 mm).

- **Condenser** (0.2 μF) across the points absorbs the self-induced primary emf (≈ 250 V) ⇒ prevents arcing at points & speeds up current collapse (higher secondary voltage).

$$V_2 = V_1 \times (N_2 / N_1) \text{ (turns ratio} \approx 100 : 1)$$

- A **ballast resistor** limits current at low speed & is bypassed at starting.
- Disadvantages : poor high speed spark (current has little time to build), points wear & need service; battery run-down effect at starting.

2.17 Magneto Ignition System



Rotating magnet cuts the coil → flux changes → emf in primary;
when the points open the primary current collapses → HV in secondary.

- A **permanent magnet** (rotating) or rotating armature generates the emf itself ⇒ **no battery**. It combines the generator, coil, breaker, condenser and distributor.
- **Types** : rotating armature, rotating magnet, polar inductor type. Used in motorcycles, scooters, aircraft & small engines (reliable at high speeds).

Point	Battery ignition	Magneto ignition
Battery	needed	not needed

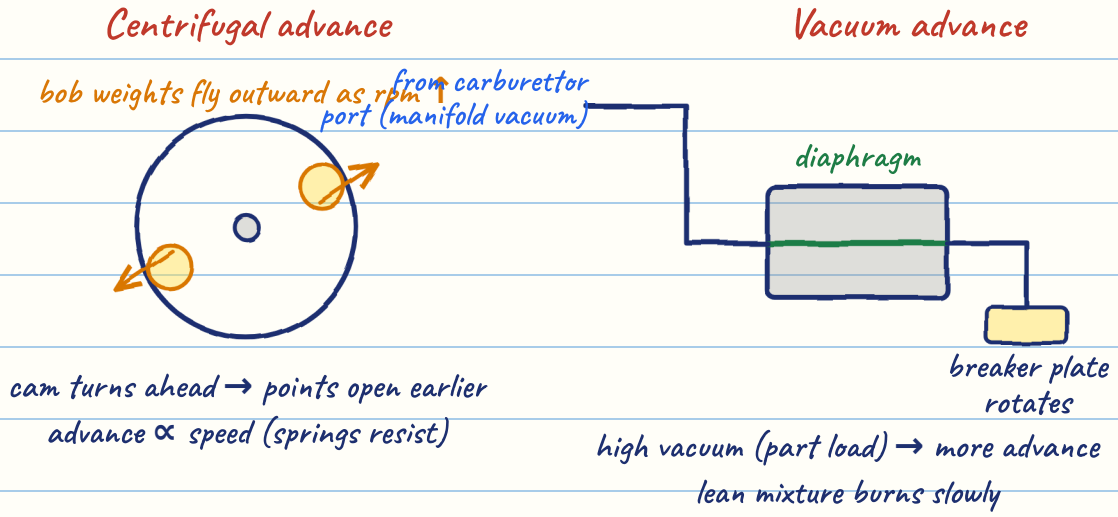
Point	Battery ignition	Magneto ignition
Starting	good spark at cranking	poor at low (cranking) speed
High speed	spark weak	very good
Maintenance	more (battery)	less
Use	cars	2-wheelers, aircraft

2.18 Electronic Ignition Systems

- **(a) Transistorised / contact-breaker assisted** : the old points carry only a small (≈ 0.5 A) current to switch a power transistor, which breaks the heavy primary current ⇒ points last longer, secondary voltage higher.
- **(b) Contact-less (triggered) type** : the breaker is replaced by a trigger device : **magnetic pick-up (reluctor)**, **Hall-effect sensor** or **optical** sensor on the distributor shaft; the **igniter / ECU module** switches primary current.
- **(c) Capacitor discharge (CDI)** : capacitor charged to ≈ 400 V, discharged through the primary by a thyristor (SCR) ⇒ very fast rise of secondary voltage (3-10 µs), good for 2-stroke & high rpm.
- **Advantages** : no mechanical wear, constant spark energy at all speeds, easier starting, better emission, accurate **timing** (even distributor-less DIS / coil-on-plug).

2.19 Spark Advance & Retard Mechanisms

- Burning needs **time** (≈ 1 ms delay) so spark must be given **before TDC (advance)**. As speed rises, the crank turns more in the same time ⇒ more advance required; at light load mixture is weak & slower ⇒ more advance; at high load, less.

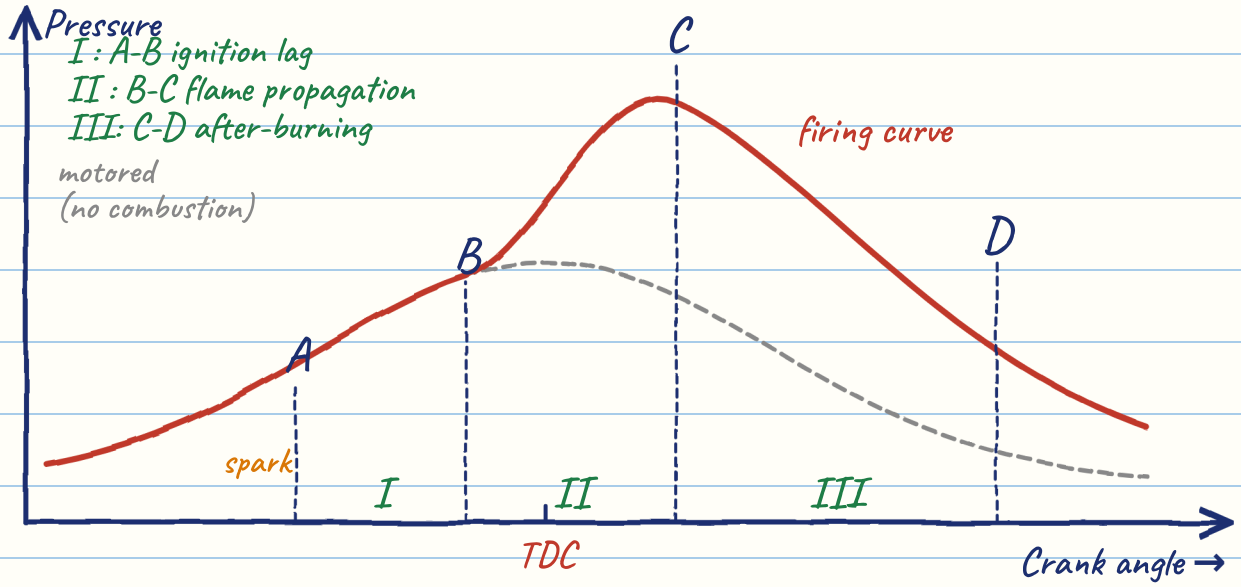


- **Centrifugal advance** (speed) : bob weights on the distributor shaft fly outward against springs → rotate the cam forward relative to shaft ⇒ points open earlier.
- **Vacuum advance** (load) : manifold vacuum on a diaphragm turns the breaker plate against cam rotation ⇒ advance at part load (high vacuum); retarded at full throttle (low vacuum).
- **Retard** : spark is delayed (set manually in old engines) for starting, to avoid kickback & knocking at low speeds. Octane selectors in some engines. Modern ECUs use a **map** with knock-sensor feedback.

UNIT - III : FUELS AND COMBUSTION

3.1 Combustion in S.I. Engines - Normal Combustion

- In SI engine a **homogeneous** air-fuel mixture is compressed and ignited by a **spark** at ≈ 10-35° before TDC. A flame front starts at the plug & travels through the charge at 10-30 m/s ⇒ turbulent flame propagation.
- Normal combustion = smooth, progressive burning of the whole charge from the spark plug to the far wall without any abnormal event. Pressure rise is ≈ 2-3 bar per degree.



Three stages of combustion

- **Stage I : Ignition lag (A-B)** - spark to first measurable pressure rise; a small nucleus of flame forms (chemical reaction, depends on temperature, pressure, mixture strength, residual gas). ≈ 10% of mass burned. **Pressure not visibly affected.**
- **Stage II : Flame propagation (B-C)** - flame front spreads through the charge; about 90% burned ⇒ rapid pressure rise to the peak, which should occur ≈ 10-15° **after TDC** for maximum power.

- **Stage III : After-burning (C-D)** - burning of the remaining charge in the corners, piston-crown boundary layers, during expansion; pressure falls (little work).

Spark timing is set so that peak pressure occurs at ≈ 10-15° after TDC (MBT - minimum advance for best torque).

3.2 Flame Speed & Importance

- **Flame speed** = rate at which the flame front advances (m/s) through the unburnt mixture. Higher flame speed ⇒ shorter combustion time ⇒ less heat loss, less chance for knock, higher efficiency & permits higher engine speed.
- Flame speed is raised by **turbulence** (swirl, squish, shape of chamber, high rpm) - laminar flame speed of petrol-air ≈ 0.3-0.5 m/s only, turbulent 10-30 m/s.

Effect of engine variables on flame speed / combustion

Variable	Effect
Air-fuel ratio	max. flame speed at slightly rich (≈ 12:1 - 13:1); lean/too rich ⇒ slow
Compression ratio	↑ CR ⇒ ↑ temp. & pressure ⇒ ↑ flame speed, ↑ efficiency
Engine speed	↑ speed ⇒ ↑ turbulence ⇒ flame speed ∝ rpm (time in ms const.)
Spark advance	too early ⇒ negative work, knock; too late ⇒ low peak pressure
Load	↑ load ⇒ less residual gas, ↑ pressure ⇒ ↑ speed, less lag
Chamber shape	compact, central plug ⇒ short flame path, high turbulence

Variable	Effect
Inlet temperature	↑ temp ⇒ ↑ flame speed but ↓ density (↓ power) & ↑ knock
Residual gas	dilutes the charge ⇒ ↓ flame speed

3.3 Abnormal Combustion

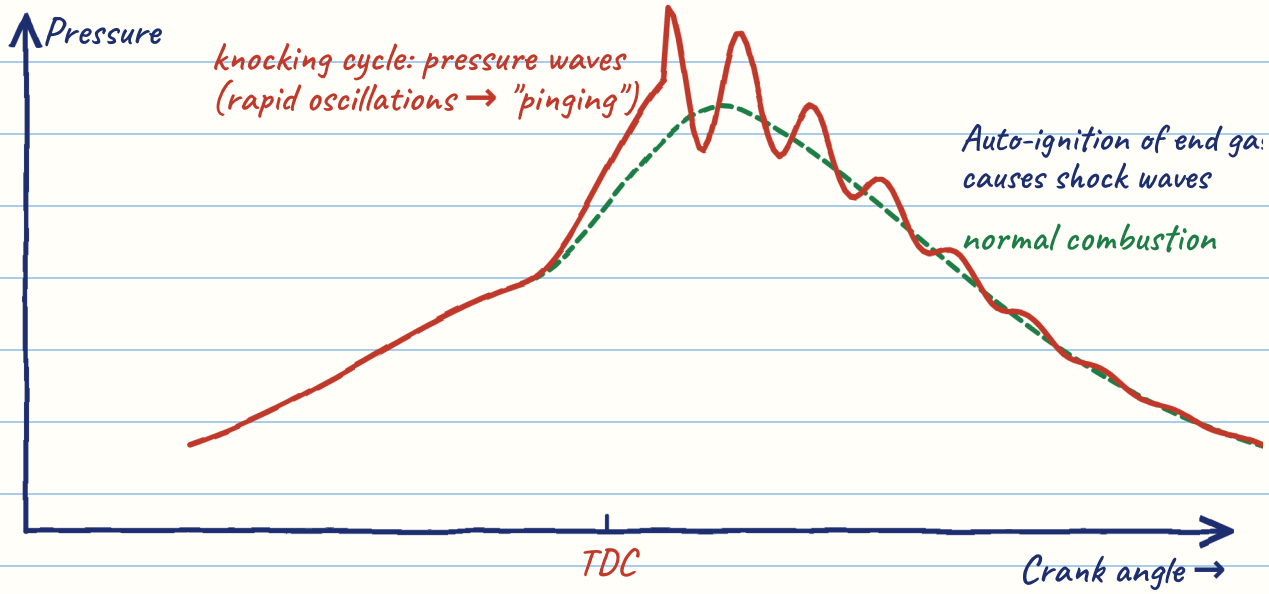
- Any combustion that **departs from the normal pattern**: (a) **Knocking (detonation)**, (b) **Pre-ignition (surface ignition)**, (c) run-on (dieseling), (d) rumble, wild ping, backfire.

(a) Pre-Ignition

- Ignition of the charge **before the spark** occurs, caused by hot spots : overheated spark-plug electrodes, exhaust valve, carbon deposits, sharp edges on the gasket.
- Effects : mixture burns too early ⇒ compression work increases (pressure rise acts against the piston) ⇒ **loss of power**, overheating, a dull thud; can cause piston damage. It can trigger knocking & vice versa (a vicious cycle).
- Remedy : use **correct heat range spark plug**, clean carbon, proper cooling, avoid sharp edges.

(b) Knocking (Detonation) in SI engines

- Normal flame compresses & heats the **end gas** (the last part of unburnt charge) ahead of the flame. If the end gas reaches its **auto-ignition temperature** and the **delay time** is shorter than the time for the flame to arrive, it burns **almost instantaneously** (≈ 1 ms) → violent pressure waves hit the cylinder wall, causing the metallic "ping / knock".



- Effects : **loss of power**, overheating of piston crown & plug, pitting / erosion of piston & head, broken rings, increased heat loss, noise, vibration, higher pollution.

Mechanism : Auto-ignition of the end gas

- Pre-flame reactions form peroxides / aldehydes in the end gas ⇒ chain branching ⇒ spontaneous ignition. High **pressure, temperature & residence time** and low octane fuel (long-chain paraffins) promote knock.

Effect of engine variables on knock

Variable	To reduce knock
Compression ratio	decrease (CR is limited by knock)
Inlet temperature & pressure	decrease (intercooler, no supercharge)
Coolant temperature	decrease (but not too low)
Spark timing	retard the spark
Mixture strength	slightly rich (≈ 10-11:1 best) or lean beyond 18:1
Engine speed	increase (more turbulence, less time)

Variable	To reduce knock
Combustion chamber	compact, central plug, short flame path, cooler end gas
Fuel	high octane number
Engine size	smaller bore (shorter flame travel)
Deposits	remove (reduce effective CR & hot spots)

Pre-ignition vs Knock

Point	Pre-ignition	Knocking
Ignition	before the spark	after spark, in end gas
Cause	hot spot	auto-ignition of end gas
Sound	dull thud	sharp metallic ping
Pressure	rises before TDC	oscillates after peak
Control	clean plugs, cooling	octane, retard, cooler

3.4 Fuel Requirements for SI engine & Fuel Rating

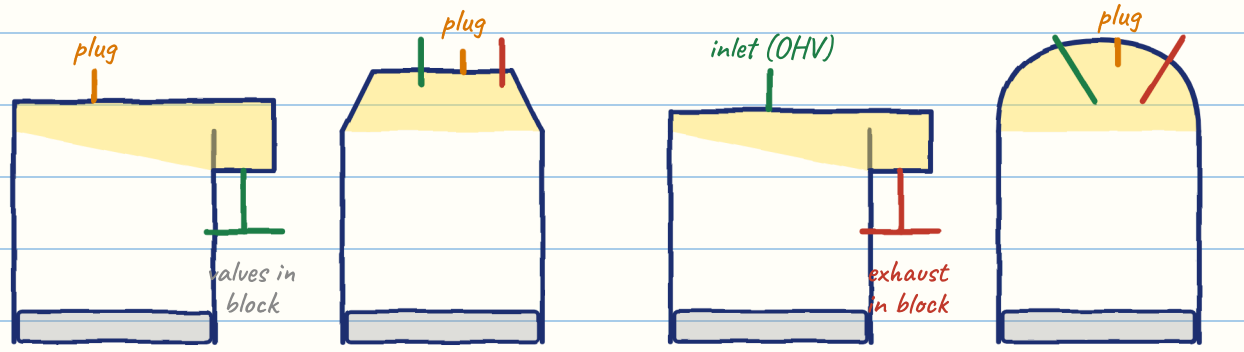
- **Ideal SI fuel** : high **anti-knock** quality (high octane), high volatility (easy starting) but not vapour-lock, high calorific value, good storage stability, low gum, low sulphur, no deposits, cheap.
- **Octane number (ON)** = % by volume of **iso-octane** in a blend of iso-octane (ON = 100) and n-heptane (ON = 0) that matches the knock intensity of the test fuel, in a standard **CFR engine** (variable-compression, single cylinder).
- **RON** (Research, 600 rpm, mild 52 °C) & **MON** (Motor, 900 rpm, 149 °C, severe) ; MON < RON; **Sensitivity** = RON − MON ; Pump rating (R+M)/2 (AKI).
- **Performance number (PN)** for fuels above 100 octane : PN = 100·(knock-limited imep of fuel / of iso-octane).

- **Highest Useful Compression Ratio (HUCR)** : max CR at which fuel runs knock-free.
- Typical: petrol 87-95 RON; LPG, CNG 105-130; ethanol 108.

3.5 Anti-Knock Additives

- **Tetra-ethyl lead (TEL)** : 0.15 - 0.8 ml/l ⇒ stops chain reaction in end gas ; **banned** (pollution, catalyst poisoning); ethylene dibromide scavenger.
- **Tetra-methyl lead**, **MMT** (manganese), **ferrocene** (iron) - metallic.
- **Oxygenates** : **MTBE**, ETBE, ethanol (E10, E20), methanol - raise ON, reduce CO.
- **Aromatics** : benzene, toluene, xylene (limited by emissions); **alkylates & isomerates** (reformed refinery blending).
- Other methods : catalytic reforming, isomerisation, alkylation, polymerisation.

3.6 SI Engine Combustion Chambers



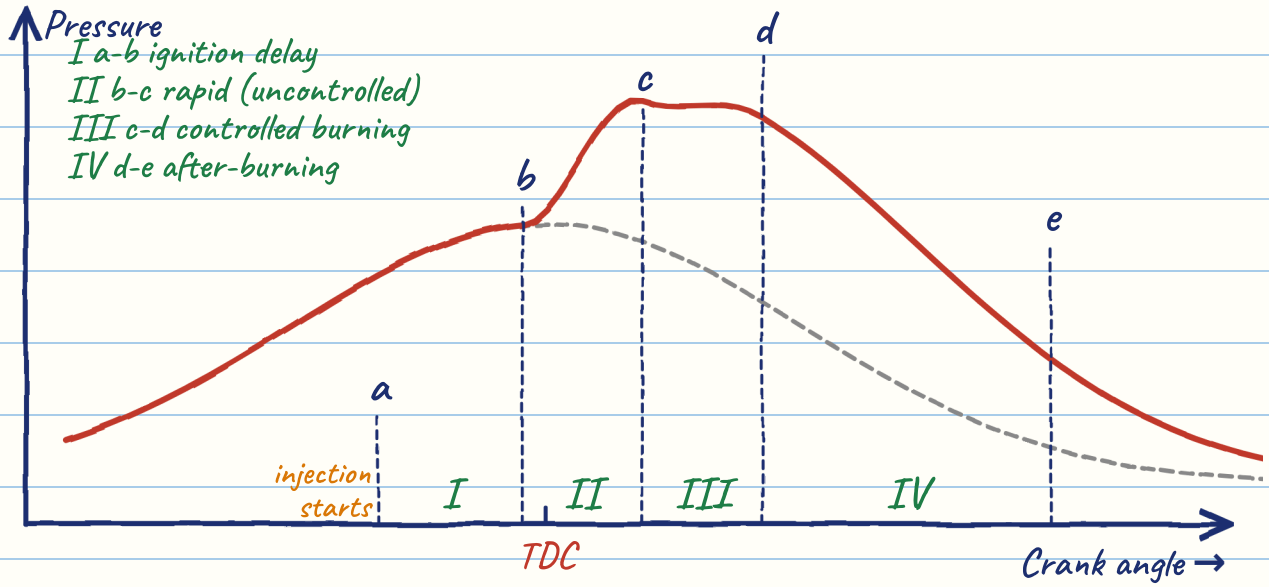
(a) L-head (b) I-head (OHV) (c) F-head (d) Hemispherical

L: simple, but low CR (<7), slow burning · I: compact, higher CR, short flame path
Hemispherical: valves inclined, central plug → shortest flame travel, big valves, best breathing

- **Requirements** : (1) high **power & thermal efficiency**, (2) low knock (compact; short flame travel), (3) high turbulence / swirl, (4) high volumetric efficiency (big valves, good breathing), (5) spark plug located centrally (or near exhaust valve - hottest part), (6) the **end gas** is cooled (large surface/volume ratio near the end), (7) quiet, low deposits, low HC emissions, easy to manufacture.
- **L-head (side valve)** : simple, long flame path; **Ricardo turbulent head** reduces the knock. **I-head / OHV** : valves above; compact, better flow ⇒ high CR. **F-head** : inlet over, exhaust in block (old Rover). **Hemispherical / pent-roof / wedge / bath-tub / Heron** : used with 4-valves per cylinder & central plug (modern).
- **Squish** : the end gas is squeezed into the chamber from the narrow clearance, giving turbulence & cooling the end gas.

3.7 Combustion in C.I. Engines

- **Heterogeneous** combustion : fuel injected (as fine spray) into the hot compressed air (≈ 600-800 °C, 30-40 bar) near the end of compression (≈ 15-25° bTDC); each droplet vaporises, mixes & **self-ignites** ⇒ burning without a spark. Overall A/F lean (20:1 to 70:1).



Four stages of combustion

- I. Ignition delay (a-b) : period between start of injection and start of burning (≈ 0.4-3 ms or 8-15° CA). Has two parts: physical delay (atomisation, heating, vaporisation, mixing) and chemical delay (pre-reactions). Pressure follows the motoring curve.
- II. Rapid (uncontrolled) combustion (b-c) : fuel accumulated during the delay burns almost at once at constant volume ⇒ steep pressure rise (≈ 8-12 bar/deg is the cause of noise).
- III. Controlled (mixing-controlled) combustion (c-d) : the fuel injected burns as it enters; the rate is controlled by the injection rate & mixing ⇒ peak pressure; almost at constant pressure.
- IV. After-burning (d-e) : residual fuel burns during expansion; incomplete burning gives smoke.

3.8 Delay Period – Importance & Effect of Engine Variables

- **Importance** : long delay ⇒ large amount of fuel accumulates ⇒ **very high rate of pressure rise** ⇒ **diesel knock**, rough running, higher peak pressure, stress & noise. Short delay ⇒ smooth, quiet engine.

Variable	Effect on delay period
Compression ratio	↑ CR ⇒ ↑ temp. & pressure ⇒ delay ↓
Intake temp. & pressure	↑ ⇒ delay ↓ (supercharging helps)
Engine speed	delay (ms) ↓ but delay in crank degrees ↑
Injection timing	optimum ≈ 10-15° bTDC; earlier ⇒ longer delay (colder air)
Fuel (cetane no.)	↑ cetane ⇒ delay ↓
Injection pressure, fine atomisation	↓ delay (better mixing)
Load (A/F)	↑ load ⇒ ↑ temp. ⇒ delay ↓
Swirl / turbulence	↑ mixing ⇒ delay ↓

3.9 Diesel Knock

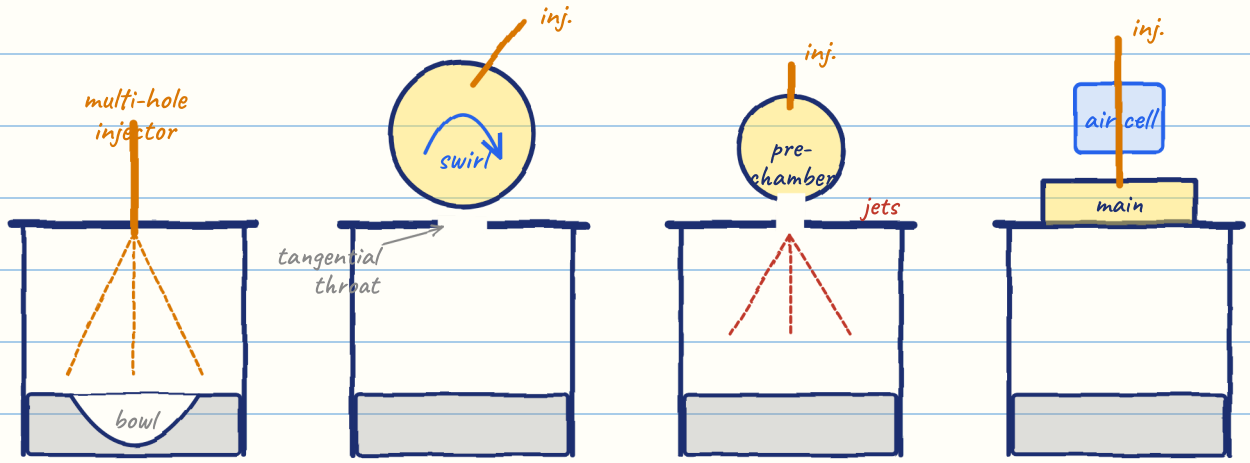
- Diesel knock occurs due to the **sudden ignition of the fuel accumulated during a long delay period** ⇒ abrupt pressure rise (≥ 10 bar/deg) with pressure waves that give the rough knocking sound.
- **Cause** : long delay (cold engine, low cetane, low CR, late injection). It happens at the **start** of combustion (in CI) while in SI knock occurs at the **end** (end gas).

SI knock vs Diesel knock

Point	SI knock	CI (diesel) knock
Where	end of combustion (end gas)	beginning of combustion
Cause	auto-ignition of end gas	too long delay, fuel accumulation
Fuel needed	high octane (high SIT)	high cetane (low SIT)
Remedy	↓ CR, retard, cooling	↑ CR, ↑ intake temp., pilot injection

- **Remedies** : use high-cetane fuel, ↑ compression ratio, ↑ intake temperature/pressure (supercharge), ↑ jacket temperature, **control injection rate** (pilot injection, pintle nozzle), good swirl, correct timing, ignition improvers (amyl-/ethyl-nitrate).

3.10 C.I. Engine Combustion Chambers



(a) Open / DI (b) Swirl chamber (c) Pre-combustion (d) Air cell

DI : high efficiency, hard cold start, noisy · IDI : quiet, smooth, but 10-15 % more fuel

(A) Direct injection (DI) chambers

- Fuel injected directly into the single open chamber in the piston crown; **multi-hole nozzle** (3-10 holes, 200-1000 bar). Types : **open (shallow depth)**, **hemispherical (Hesselman/ Saurer)**, **toroidal / re-entrant (Mexican-hat)** - give swirl by **inlet-port (helical) design**.
- Merits : highest efficiency & lowest fuel consumption, easy cold start (no glow plug), simple head. Demerits : high injection pressure, noisy, sensitive to fuel quality, high NOx.

(B) Indirect injection (IDI) / divided chambers

- Chamber divided into main chamber & a smaller **auxiliary chamber** (≈ 50-80% of clearance volume in swirl chamber); fuel is injected into the auxiliary one, partially burns, then passes through a throat into the main chamber (strong turbulence).
- **Swirl chamber (Ricardo Comet)** : spherical, tangential throat; swirl; speed up to 4500 rpm. **Pre-combustion chamber** : 25-40% volume, restricted passages; quiet. **Air-cell / energy cell** (Lanova) : air stored in the cell enters at the end of burning.
- Merits : low injection pressure (100-150 bar, single-hole pintle nozzle), smooth, quiet, uses wide range of fuels, high speed. Demerits : poor cold start (needs **glow plug**), higher heat loss ⇒ ≈ 10-15% higher fuel consumption.

3.11 Fuel Requirements & Rating of C.I. Fuels

- **Requirements** : high **cetane number** (short delay), good volatility, suitable viscosity (for pump lubrication & spray), high calorific value, low sulphur, low pour / cloud point, good oxidation stability, low carbon residue & ash, cleanliness, non-corrosive.

- **Cetane number (CN)** = % by volume of **cetane (n-hexadecane, CN = 100)** in a mixture with **α-methyl-naphthalene (CN = 0)** that matches the ignition quality of the fuel in a standard CFR engine. (Modern reference: **heptamethylnonane**, CN = 15.)
- **Diesel index / Cetane index** = calculated (API gravity, aniline point) ; typical diesel CN : 45 - 55 (BS : 40 min).
- **Relation** : fuels good for SI have low cetane number and vice versa (high SIT = high octane; low SIT = high cetane).
- **Ignition improvers** : ethyl nitrate, amyl nitrate, di-tertiary butyl peroxide (0.5-1 %).

Summary : SI → octane no., anti-knock, spark ; CI → cetane no., short delay, auto-ignition. Opposite fuel qualities!

3.12 Combustion in SI vs CI Engines

Point	SI engine	CI engine
Mixture	homogeneous	heterogeneous
Start of burning	spark (timed)	auto-ignition (after delay)
Flame	propagates from the plug	many nuclei, spray burning
Load control	quantity of mixture (throttle)	quantity of fuel (A/F varies)
Abnormality	knock (end gas), pre-ignition	diesel knock (long delay)
Fuel quality	octane number	cetane number
Max. CR limit	knock	weight / pressure / stress

Important questions (exam)

- Explain stages of combustion in SI engine with p-θ diagram.
- Differentiate pre-ignition and knocking; explain the effect of engine variables on knocking.

- What is octane number? How is it measured ? Name anti-knock additives.
- Explain the stages of combustion in a CI engine; importance of delay period.
- Explain diesel knock and compare it with SI knock.
- Describe DI and IDI combustion chambers with neat sketches. Define cetane number.

UNIT – IV : TESTING AND PERFORMANCE OF I.C. ENGINES

4.1 Why test an engine ?

- To determine **power, efficiency, fuel consumption** & heat distribution; to compare engines, to check design & after repair; for emission certification; to draw performance curves.

4.2 Parameters of Performance

- **Indicated power (IP)** : power developed in the cylinder (from indicator diagram).

$$IP = p_{im} L A n N_c / 60000 \text{ (kW)} \quad [p \text{ in } N/m^2, L \text{ in } m, A \text{ in } m^2]$$

- n = number of cylinders ; **N_c = N/2** for 4-stroke (power strokes / min), = N for 2-stroke ; N = rpm.
- **Brake power (BP)** : useful power at the crankshaft, measured by a dynamometer.

$$BP = 2\pi NT / 60000 \text{ (kW)}, \; T \text{ in } N\cdot m$$

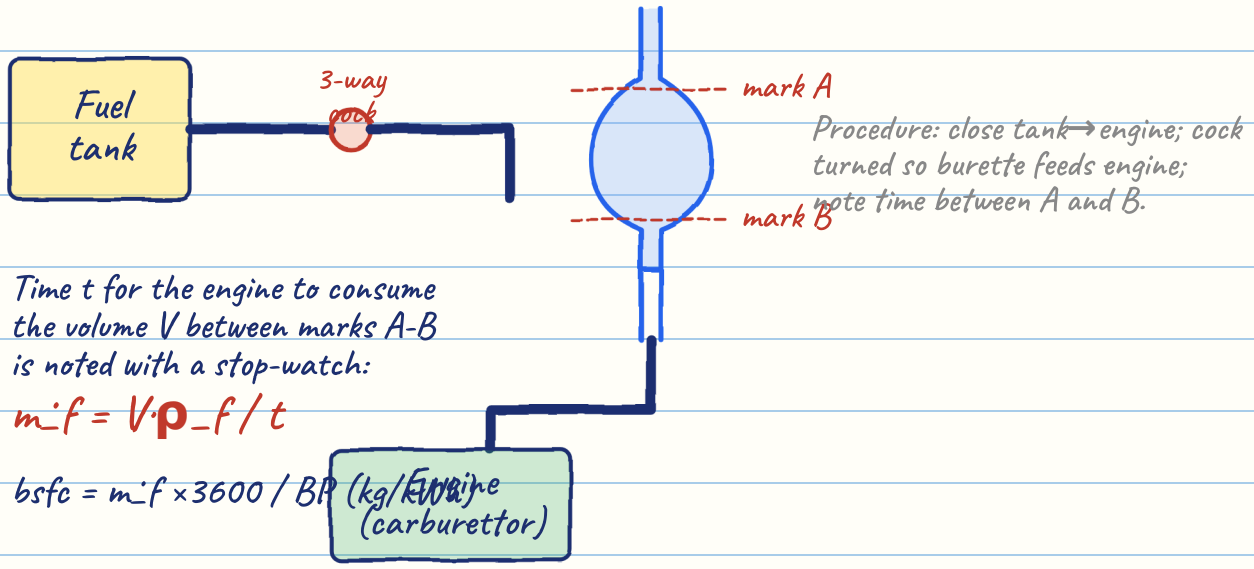
- **Friction power (FP)** = IP – BP (bearing, piston-ring, valve gear, pumping, accessories).
- **Mechanical efficiency** η_m = BP / IP = BP/(BP + FP) (≈ 80-90 %).
- **Mean effective pressure** : the hypothetical constant pressure acting on the piston for one stroke giving the same work : imep (from IP), **bmep** = BP × 60000 / (L A n N_c), **fmep** = imep – bmep.
- **Indicated thermal efficiency** η_{ith} = IP / (m_f × CV); **Brake thermal efficiency** η_{bth} = BP / (m_f × CV) ; m_f in kg/s, CV in kJ/kg.
- **Specific fuel consumption** : **bsfc** = m_f(kg/h) / BP (kW) kg/kWh; isfc similarly with IP. ηbth = 3600 / (bsfc × CV).

- **Volumetric efficiency** η_v = actual air drawn / swept volume at atm. conditions (80-90 % SI, 85-95 % CI).
- **Relative (efficiency ratio) efficiency** = η_{th}(actual) / $\eta_{air\text{-}standard}$.
- **Air-fuel ratio**, **specific output** (kW/l of swept volume), **mean piston speed**, **torque**.

4.3 Measurement of Cylinder Pressure (Indicators)

- Objective : p-V / p-θ diagram ⇒ imep, IP. **Mechanical indicators** (Watt / Crosby / Dobbie-McInnes) work only at low speed because of inertia.
- **Electronic indicators** : the pressure transducer in the cylinder head (flush-mounted) + charge amplifier + crank-angle encoder + oscilloscope / DAQ.
- Types : **Piezo-electric** (quartz - best; high natural frequency > 100 kHz), strain gauge, capacitive, optical, magnetic (inductance) pick-up.
- The p-θ diagram is converted to p-V using the slider-crank relation; **TDC detection** is critical (error of 1° gives ~ 3-5 % error in imep).

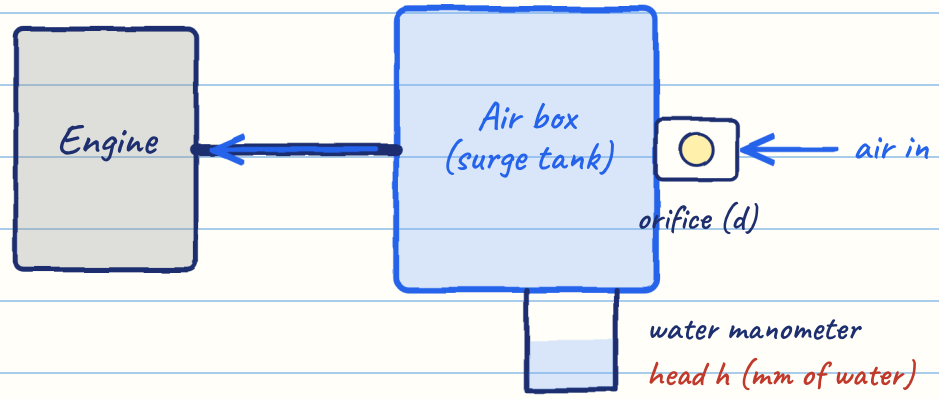
4.4 Measurement of Fuel Consumption



- (a) **Volumetric method** : burette with 2 marks (25, 50, 100 cc) ; time t for known volume V is noted ⇒ $\dot{m}_f = V\rho / t$. Simple, accurate for small engines.
- (b) **Gravimetric method** : fuel tank on a weighing balance, time for a fixed mass (e.g. 50 g) consumed. Independent of temperature (density).
- (c) **Flow meters** : positive displacement (PD), Coriolis, turbine; continuous reading (for large engines & ECU testing).
- For **diesel**, injector return line must be returned downstream of the measurement.

4.5 Measurement of Air Intake

Air-box method of air-flow measurement



Volume flow $V = C_d \cdot (\pi/4)\, d^2 \cdot \sqrt{(2\, g\, h_a)}\ (m^3/s)$, $h_a = h \cdot \rho_w / \rho_a$

Box damps pulsations of the engine suction; box volume ≈ 500 × swept volume.

- (a) Air-box with orifice (above) : pulsation damped; $V = Cd\ (\pi/4) d^2 \sqrt{(2\, g\, h_a)}$; typical $Cd = 0.6$; $m_a = \rho_a V$.
- (b) Viscous-flow (Alcock) meter : laminar flow through a honeycomb element ⇒ $\Delta p \propto$ flow (linear) ⇒ no damping tank required.
- (c) Hot-wire (MAF) sensors, turbine / vane meters, thermal mass flow - used in vehicles & modern tests.
- (d) Pitot / nozzle & indirect : from the A/F ratio using exhaust analysis (carbon balance method).

4.6 Measurement of Exhaust Gas Composition

- Purpose : combustion quality, A/F ratio, emissions (CO, HC, NOx, CO_2, O_2, smoke).
- Orsat apparatus (chemical / absorption) : CO_2 (KOH), O_2 (alkaline pyrogallol), CO (cuprous chloride) absorbed in order; dry analysis.

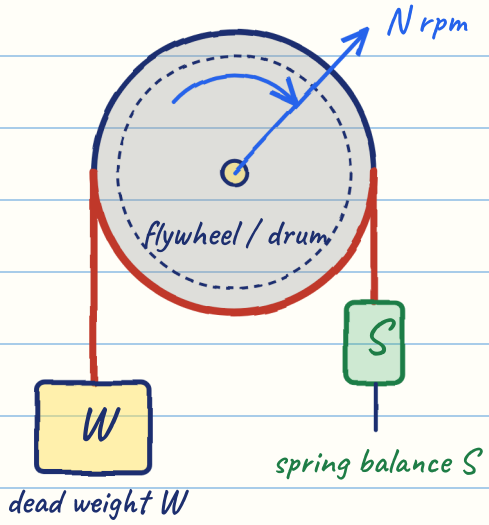
- **NDIR** (non-dispersive infrared) : CO, CO_2, HC. **FID** (flame ionisation) : total hydro-carbons. **Chemiluminescence (CLD)** : NO/NOx. **Paramagnetic** analyser / **zirconia lambda sensor** : O_2. **Gas chromatograph** : species.
- **Smoke meters** : Bosch (filter paper darkness, FSN), Hartridge (light extinction), opacimeter ; **PM** : gravimetric (dilution tunnel).
- **A/F from exhaust** : lean ⇒ high O_2; rich ⇒ high CO; stoichiometric ⇒ CO ≈ 0.

4.7 Measurement of Brake Power

- **Dynamometer** = device that **absorbs & measures** the engine power. Classified : **absorption** (all power converted to heat: rope brake, Prony brake, hydraulic, eddy-current, fan, electric (generating)) and **transmission** (power passes to a load : belt, epicyclic train, torsion dynamometers).

(a) Rope brake dynamometer

Rope brake dynamometer



Effective radius $R = (D + d)/2$

Net brake load = $(W - S)$

Torque $T = (W - S) \cdot R$ (N·m)

$BP = 2\pi NT / 60000$ (kW)

W, S in N (kg × 9.81)

D = drum dia, d = rope dia

Needs cooling water in the drum (heat = friction work)

- Simple & cheap, used for small engines. Drum is water-cooled. Rope turns around drum, with a dead weight on one end & spring balance on the other. **Disadvantage** : rope wears, load fluctuates, not for high power.

(b) Prony brake

- Two wooden **brake blocks** clamp over the flywheel through a bolt & spring; a **lever arm** (length l) carries a weight W (or load cell). $T = W \times l$, $BP = 2\pi NWl / 60000$. Adjusting the nuts changes the friction. Bad for high power as friction coefficient is unstable (chattering).

(c) Hydraulic (water-brake / Froude) dynamometer

- Rotor with pockets rotates in a casing filled with water; the **shearing & vortex flow** of water gives the resisting torque; torque arm measures reaction torque on the casing. Load regulated by water level/sluice. Handles large powers (up to many MW).

(d) Eddy-current dynamometer

- A steel/copper disc rotating between electromagnets (DC excited) generates **eddy currents** $\Rightarrow$ braking torque $\propto$ field current; the reaction measured by load cell. Quick response, fine control, widely used in modern test beds.

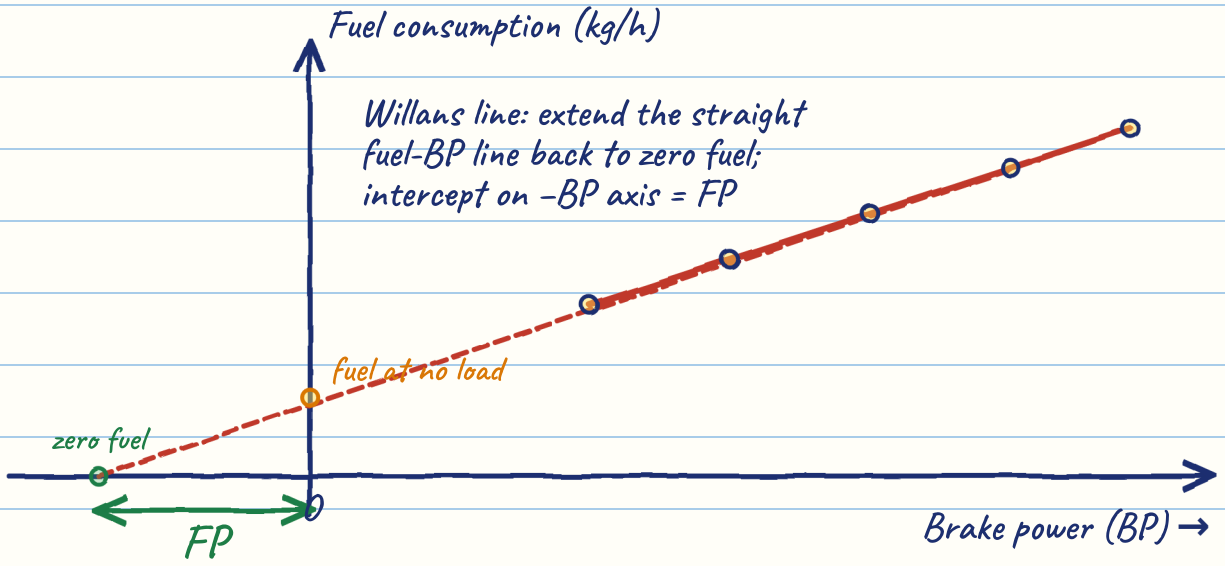
(e) Electric (swinging-field) dynamometer

- DC / AC generator with cradled stator; can both **absorb & drive (motoring)** the engine $\Rightarrow$ motoring test possible.

4.8 Determination of Frictional Power & Indicated Power

- **(1) Indicator diagram** - IP from imep (area / length × spring constant).

- (2) **Willans line method** (fuel-rack method) - used for **diesel engines** (smoke limit); at constant speed plot fuel consumption vs BP; extrapolate the (approximately straight) line back to **zero fuel** ⇒ intercept = −FP.



(3) Morse test (for multi-cylinder SI engines)

- The engine runs at constant speed at full throttle, BP measured. Then **one cylinder is cut out** (spark shorted) & the load reduced to bring the speed back to the same value; BP measured again. The reduction is the **IP of that cylinder.**

$$IP_1 = B - B_1,\ IP_2 = B - B_2,\ \ldots;\ IP = \Sigma IP_i;\ FP = IP - B$$

- Assumption : FP is the same with all cylinders firing or not.

Example (4-cyl) : B = 30 kW; B_1 = 20, B_2 = 19.5, B_3 = 20.5, B_4 = 19 kW ⇒ IP = 10 + 10.5 + 9.5 + 11 = **41 kW**; FP = 41 − 30 = **11 kW**; η_m = 30/41 = **73.2 %**.

(4) Motoring test

- The engine (fuelled off, hot) is **driven by an electric dynamometer** at rated speed; power absorbed = FP. Gives pumping + friction (compression work included) for all type engines. Drawback: the temperatures, pressure, & lube oil conditions differ from firing.

(5) Retardation (deceleration) test

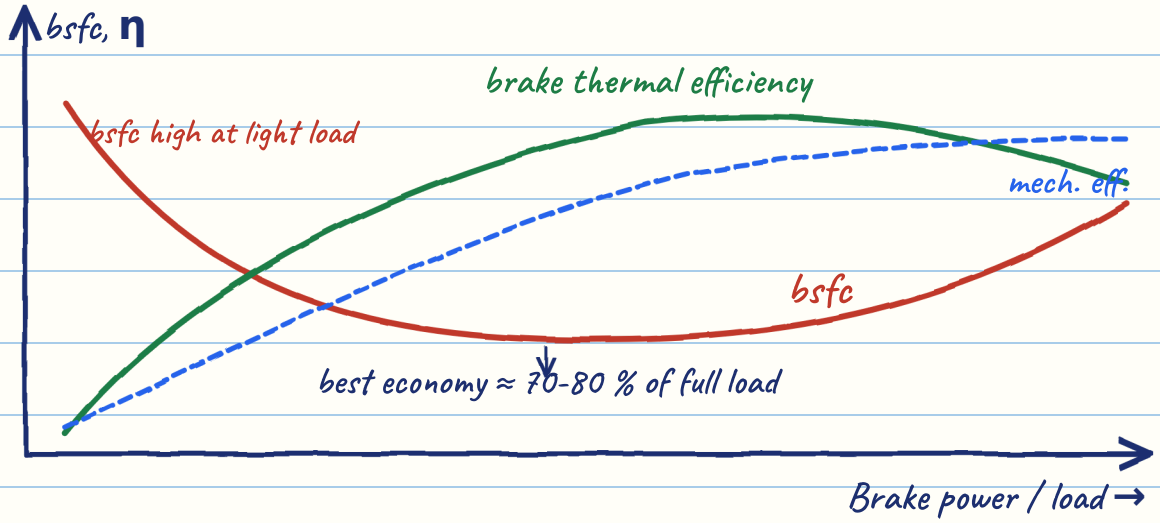
- Engine at rated speed, **cut-off** (ignition/fuel off) and the speed fall measured with time: with & without a load. FP ∝ moment of inertia × (dN/dt). Requires I (flywheel).

$$FP = (2\pi/60)^2 I N (dN/dt) / 1000$$

- **Comparison** : Morse test only for multi-cylinder SI ; Willans line for CI ; motoring for all (with electric dynamometer).

4.9 Performance Test

- Conducted to find BP, bsfc, η_m, η_{th}, η_v & draw performance curves. Types :
- (i) **Variable load test at constant speed** (e.g. generating set): load varies; measure BP, fuel, air, temperatures.
- (ii) **Variable speed at constant (full) load (throttle)** : find torque, power & bsfc vs speed - vehicle engines.
- (iii) **Morse test** & (iv) **Heat balance test** at rated speed.
- Observations : speed, load (weights, spring balance), fuel time, cooling-water flow & temperatures, exhaust temperature, air flow (manometer), ambient conditions.



- bsfc is **high at light loads** because of friction share; minimum at ≈ 70-80 % load; slightly increases near full load due to rich mixture. ηm increases with load (FP almost constant).

4.10 Heat Balance Sheet & Chart

- Distribution of the **heat supplied by the fuel** ($Q_s = m_f \times CV$) :

$$Q_s = BP + Q_w + Q_g + Q_{unacc.}$$

- (1) **Heat to BP** (kW), (2) **heat to cooling water** $Q_w = m_w c_w (T_{wo} - T_{wi})$, (3) **heat to exhaust gas** $Q_g = m_g c_{pg} (T_g - T_a)$ ($m_g = m_a + m_f$), (4) **unaccounted** (radiation, convection, friction, incomplete combustion, oil cooling).

Item	Typical SI	Typical CI
Brake power	25 - 30 %	30 - 38 %
Cooling water & oil	25 - 30 %	25 - 30 %
Exhaust gas	30 - 40 %	25 - 35 %
Unaccounted	5 - 10 %	5 - 8 %

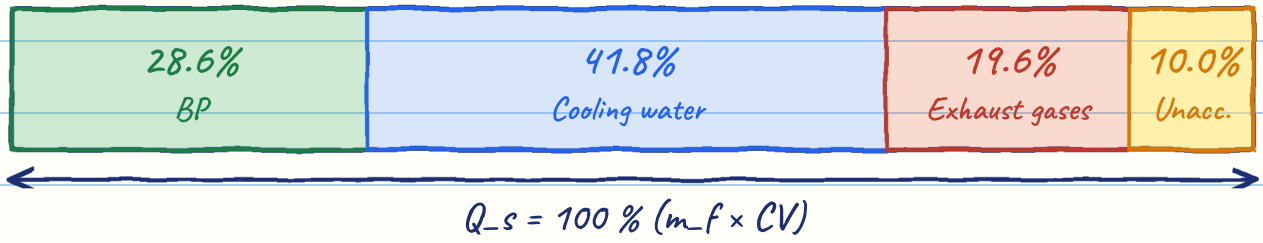
Heat balance sheet (kW and %) - Example

- 4-stroke diesel: BP = 8 kW, fuel = 2.4 kg/h (CV = 42000 kJ/kg), cooling water 0.08 kg/s with 35 K rise, air = 48 kg/h, exhaust temp 400 °C, ambient 27 °C, cp(gas) = 1.05 kJ/kg K.

- Heat supplied $Q_s = 2.4 \times 42000 / 3600 = 28$ kW (100 %).

Heat in	kW	%
Brake power	8.00	28.6
Cooling water (0.08 × 4.18 × 35)	11.70	41.8
Exhaust (50.4/3600 × 1.05 × 373)	5.48	19.6
Unaccounted (by difference)	2.82	10.0
Total	28.00	100

Heat balance chart (heat supplied by fuel = 100 %)



BP = useful work (28.6 %)
Cooling water = $m_w c_w (T_{out} - T_{in})$
Exhaust gas = $m_g c_{pg} (T_g - T_a)$
Unaccounted = Q_s − (BP + water + exhaust) → radiation, friction, etc.

4.11 Solved Problems

- **Problem 1** (BP by rope brake): drum $D = 0.6$ m, rope $d = 0.02$ m, $W = 40$ kg, $S = 5$ kg, $N = 600$ rpm.
  → $R = (0.6 + 0.02)/2 = 0.31$ m ; $(W - S) = 35 \times 9.81 = 343.35$ N ; $T = 343.35 \times 0.31 = 106.4$ N·m
  → $BP = 2\pi \times 600 \times 106.4 / 60000 = 6.69$ kW

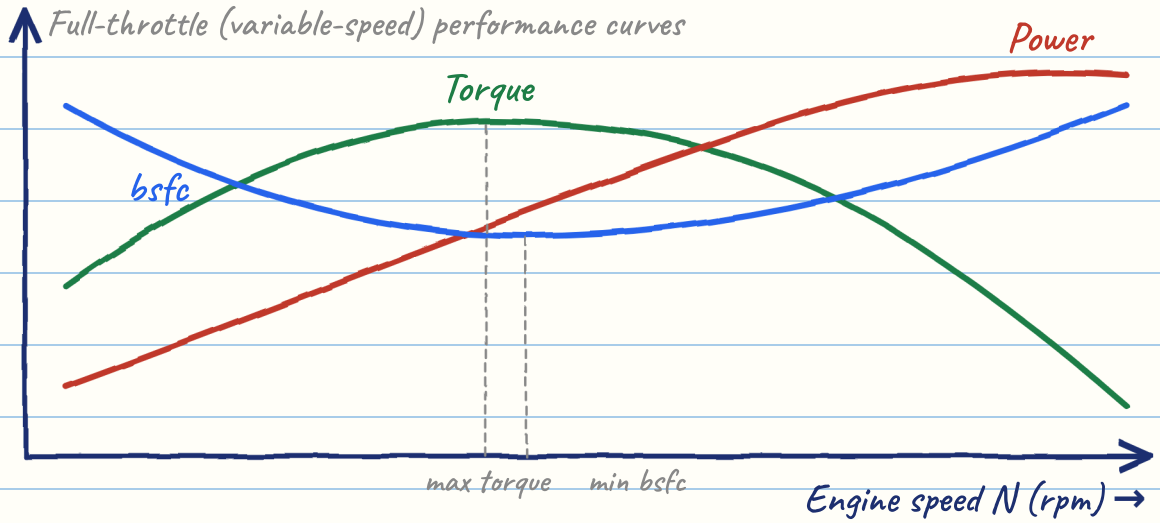
- **Problem 2** (IP from imep): 4-stroke, D = 100 mm, L = 150 mm, N = 1200 rpm, imep = 6 bar, 1 cylinder.
  → $A = (\pi/4)(0.1)^2 = 7.854\times10^{-3}\ m^2$; $IP = 6\times10^5 \times 0.15 \times 7.854\times10^{-3} \times (1200/2) / 60000 =$ **7.07 kW**
- **Problem 3** : the engine of Problem 2 has BP = 5.6 kW, uses 2.2 kg/h fuel of CV = 44000 kJ/kg.
  → $\eta_m = 5.6/7.07 =$ **79.2 %**; bsfc = 2.2/5.6 = **0.393 kg/kWh**
  → $\eta_{bth} = 5.6/(2.2 \times 44000/3600) = 5.6/26.9 =$ **20.8 %**; $\eta_{ith} = 7.07/26.9 =$ **26.3 %**
- **Problem 4** : bmep of a 4-cyl engine: BP = 30 kW, D = 90 mm, L = 100 mm, N = 3000 rpm.
  → $A = 6.362\times10^{-3}\ m^2$; $N_c = 3000/2 = 1500$; $bmep = 30 \times 60000 / (0.1 \times 6.362\times10^{-3} \times 4 \times 1500) = 4.72\times10^5\ N/m^2 =$ **4.72 bar**

> Remember : 1 bar = $10^5\ N/m^2$ = 100 kPa ; BP formula uses N in rpm ; $N_c = N/2$ (4-S).

4.12 Other Measurements in a Test Rig

- **Speed** : mechanical / digital **tachometer**, stroboscope, magnetic or optical pick-up (pulse counter), 60-tooth wheel.
- **Temperature** : thermocouples (K-type exhaust, J-type) , RTD (Pt-100) for water & oil; **pressure** : Bourdon gauge (oil), manometers (air).
- **Cooling-water flow** : rotameter / volumetric tank & stopwatch; **oil** : by sump gauge.
- **Calorific value of fuel** : bomb calorimeter (solid / liquid) , Junker's gas calorimeter (gas).

4.13 Variable-speed Performance Curves



- **Torque** rises with speed to a maximum (≈ 50-60 % of rated rpm, best filling) and then falls (volumetric efficiency & mechanical efficiency fall).
- **Power** increases almost linearly then flattens near rated speed (rises until torque falls faster than N rises); beyond rated speed power falls.
- **bsfc** is a "bathtub" curve, minimum at moderate speed (near max torque).
- Engine **flexibility** = ratio of speeds at max power to max torque; a flat torque curve = better vehicle driveability.

4.14 Formula Sheet

Quantity	Formula
IP	$p_{im} L A n (N/2) / 60000$ kW
BP	$2\pi NT/60000 = (W-S)\pi D_e N/60000$ kW
FP	IP − BP
η_m	BP / IP
bmep	BP × 60000 / (L A n N_c)
bsfc	m_f (kg/h) / BP (kW)

Quantity	Formula
η_{bth}	$BP / (\dot{m}_f \cdot CV)$, or $3600/(bsfc \cdot CV)$
η_v	V_{actual} / V_{swept}
Morse test	$IP_i = B - B_i$
Heat supplied	$\dot{m}_f \times CV$

4.15 Important Questions

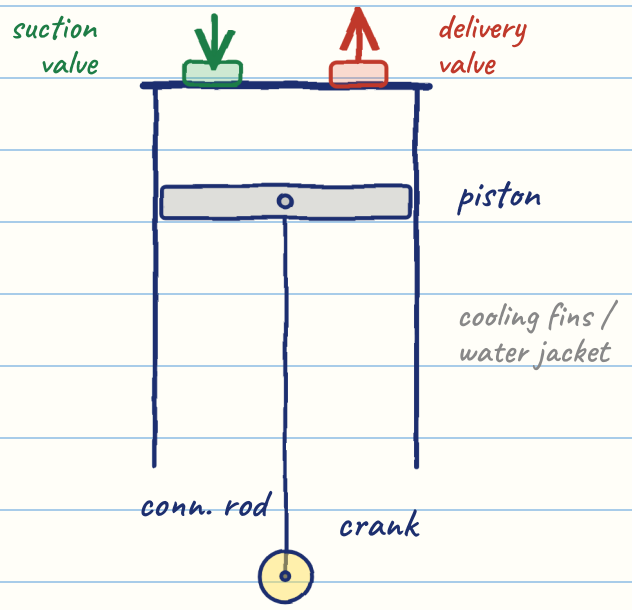
- Define IP, BP, FP, bmep, bsfc, mechanical & thermal efficiency.
- Explain the Morse test & Willans-line method for finding the friction power.
- Describe the measurement of brake power using a rope brake / hydraulic dynamometer.
- Explain how air consumption & fuel consumption are measured in the laboratory.
- Prepare a heat-balance sheet; explain the chart & its significance.

UNIT – V : AIR COMPRESSORS

5.1 Introduction & Classification

- **Compressor** : a machine that **raises the pressure of a gas (air)** by doing work on it - mechanical energy is converted into pressure energy. Uses : pneumatic tools, drilling, spray painting, air conditioning, IC-engine supercharging, gas turbines, refrigeration.
- By pressure ratio $r_p = p_2/p_1$: **Fan** ($r_p < 1.1$), **Blower** (1.1 - 4), **Compressor** (> 4).
- **(a) Positive displacement** : volume of trapped air is reduced mechanically.
  → **Reciprocating** (piston-cylinder) ; **Rotary** : Roots blower, vane (sliding vane), screw, lobe, scroll.
- **(b) Dynamic (continuous flow)** : kinetic energy given to the air is converted into pressure in a diffuser : **Centrifugal** (radial) and **Axial-flow** compressors.
- Reciprocating = high pressure, small flow ; rotary = medium ; dynamic = very large flow, moderate pressure.

5.2 Reciprocating Compressor



Suction stroke : piston ↓, suction valve opens, air enters.

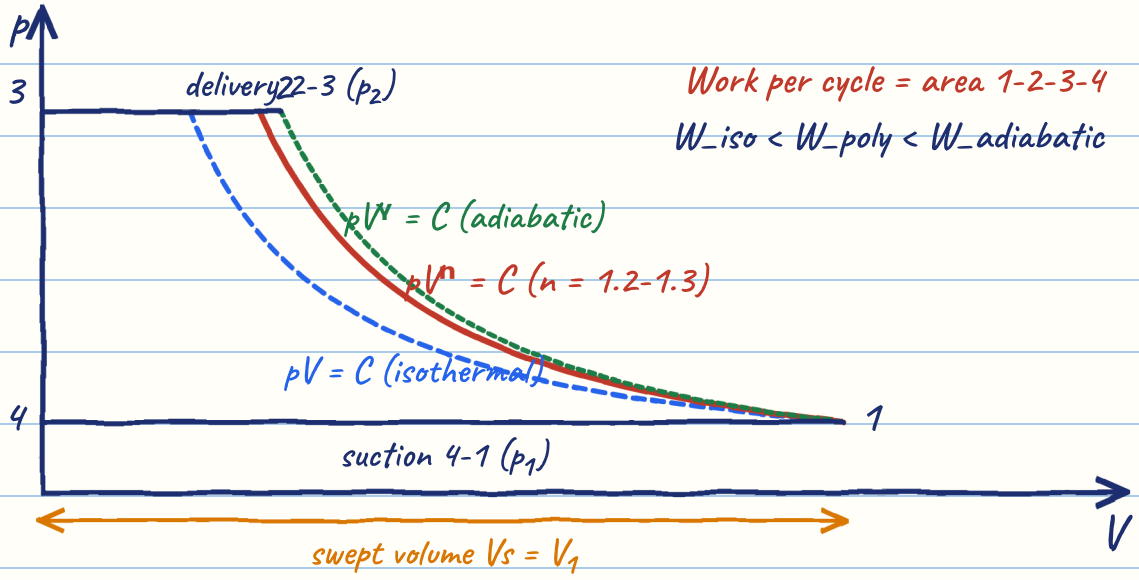
Compression : both valves shut, air compressed (p ↑, V ↓).

Delivery : delivery valve opens when p = receiver pressure.

Valves are automatic (spring-loaded plate / reed) - opened by Δp.

- **Parts** : cylinder, piston, connecting rod, crank, **suction & delivery valves** (spring-loaded, automatic), cooling jacket / fins, air receiver, filter, lubrication system, safety valve.
- May be **single acting** (one face) or **double acting** (both faces), single or multi-stage, vertical / horizontal / V / W arrangement.
- **Working** : (1) suction at p_1 with delivery valve closed, (2) compression - both valves closed, p rises, (3) **delivery** when $p = p_2$ (receiver pressure) the delivery valve opens & air is pushed out at constant p_2, (4) piston at end, clearance air remains.

5.3 Work of Compression – no clearance



- The cycle is 4-1 (suction), 1-2 (compression), 2-3 (delivery), 3-4 (pressure drop). Work per cycle = area enclosed = $\int V\,dp$.
- **Isothermal** ($pV = C$; perfect cooling):

$$W_{iso} = p_1 V_1 \ln(p_2/p_1)$$

- **Polytropic** ($pV^n = C$, $n = 1.2\text{-}1.35$):

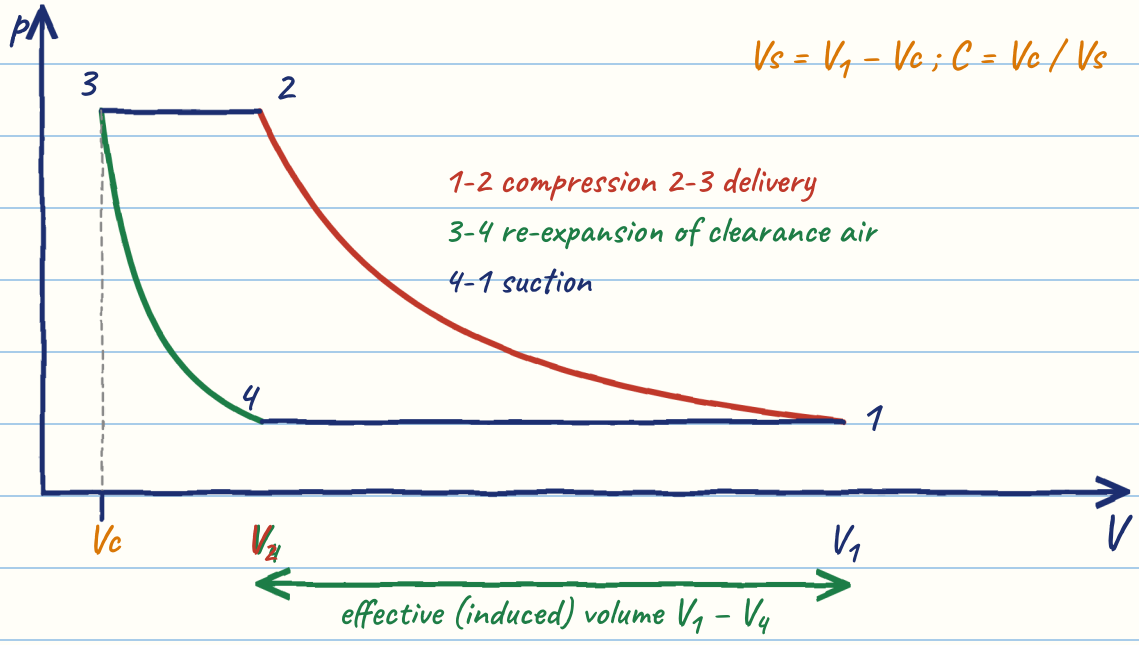
$$W = [n/(n-1)]\, p_1 V_1 [(p_2/p_1)^{(n-1)/n} - 1]$$

- **Adiabatic** ($n = \gamma = 1.4$, no cooling): same with γ in place of $n \Rightarrow$ **maximum work**.
- Delivery temperature: $T_2 = T_1 (p_2/p_1)^{(n-1)/n}$.

$W_{iso} < W_{poly} < W_{adiabatic} \Rightarrow$ cool the cylinder as much as possible (jackets, fins, intercooler) to approach isothermal.

- **Power** (single-acting, single cylinder): $P = W \times N / 60$ (W per cycle, N rpm); double acting $\Rightarrow \times 2$.
- **Isothermal efficiency** $\eta_{iso} = W_{iso} / W_{actual\ (indicated)}$ (typically 65-80 %).
- **Mechanical efficiency** $\eta_m = IP$ / shaft power (80-90 %).

5.4 Effect of Clearance Volume



$V_s = V_1 - V_c$; $C = V_c / V_s$

- A clearance volume V_c (3-10 % of V_s) is needed so the piston does not strike the head and valves can operate. After delivery, the clearance air at p_2 **re-expands** (3-4) to p_1 before the suction valve can open ⇒ part of the stroke is lost.
- **Effective (induced) volume = $V_1 - V_4$** ; swept volume $V_s = V_1 - V_c$; **clearance ratio $C = V_c / V_s$.**
- With polytropic re-expansion $V_4 = V_c (p_2/p_1)^{1/n}$:

$$\eta_v = (V_1 - V_4)/V_s = 1 + C - C\,(p_2/p_1)^{1/n}$$

- **Effects of clearance : (1) reduces the volumetric efficiency & capacity** ($\eta_v \downarrow$ with $r_p \uparrow$ and $C \uparrow$); (2) **no effect on work per kg of air** (work of compression and re-expansion cancels for same n); (3) at very high r_p, $\eta_v = 0$: $(p_2/p_1)_{max} = (1 + 1/C)^n$ ⇒ the reason for multi-staging; for $C = 0.05$, $n = 1.3 \rightarrow r_{max} = 52$.

5.5 Volumetric Efficiency & FAD

- **Volumetric efficiency** η_v = volume of free air induced / swept volume (clearance, valve losses, heating of the intake air, leakage past rings & valves, pressure drop in the suction valve lower it). Typically 60-90 %.
- **Overall (free-air) volumetric efficiency** η_{vo} = FAD per stroke / V_s (reduce to atmospheric p, T).
- **FAD (free air delivery)** = volume of air delivered reduced to atmospheric (intake) conditions : $V_{FAD} = p\,V\,T_a / (p_a\,T)$.
- **Compressor capacity** = FAD per unit time. **Indicated power** from indicator diagram.

Worked example 1

- Single-acting single-stage compressor, bore 150 mm, stroke 200 mm, N = 300 rpm, p_1 = 1 bar, p_2 = 7 bar, n = 1.3, clearance 5 % of swept volume. Find η_v, FAD (induced volume/min), IP and η_{iso}.

  → $V_s = (\pi/4)(0.15)^2(0.2) = 3.534\times10^{-3}\ m^3$

  → $\eta_v = 1 + 0.05 - 0.05\,(7)^{1/1.3} = 1.05 - 0.05(4.467) =$ **0.827**

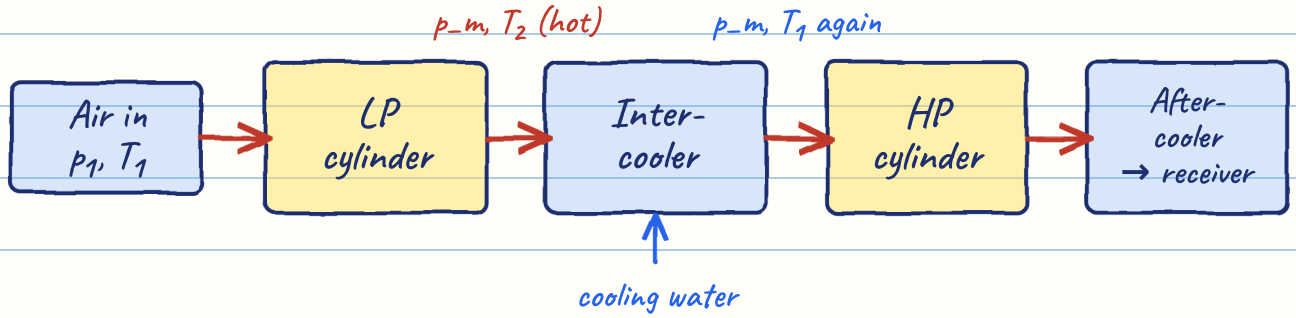
  → Induced volume per cycle = $0.827 \times 3.534\times10^{-3} = 2.921\times10^{-3}\ m^3$; per minute = **0.876 m^3/min**

  → $W = (1.3/0.3) \times 10^5 \times 2.921\times10^{-3} \times [7^{0.3/1.3} - 1] = 4.333 \times 292.1 \times 0.567 =$ **717.6 J/cycle**

  → $IP = 717.6 \times 300/60 =$ **3.59 kW**; $W_{iso} = 10^5 \times 2.921\times10^{-3} \times \ln 7 = 568.5$ J $\Rightarrow \eta_{iso} = 568.5/717.6 =$ **79.2 %**

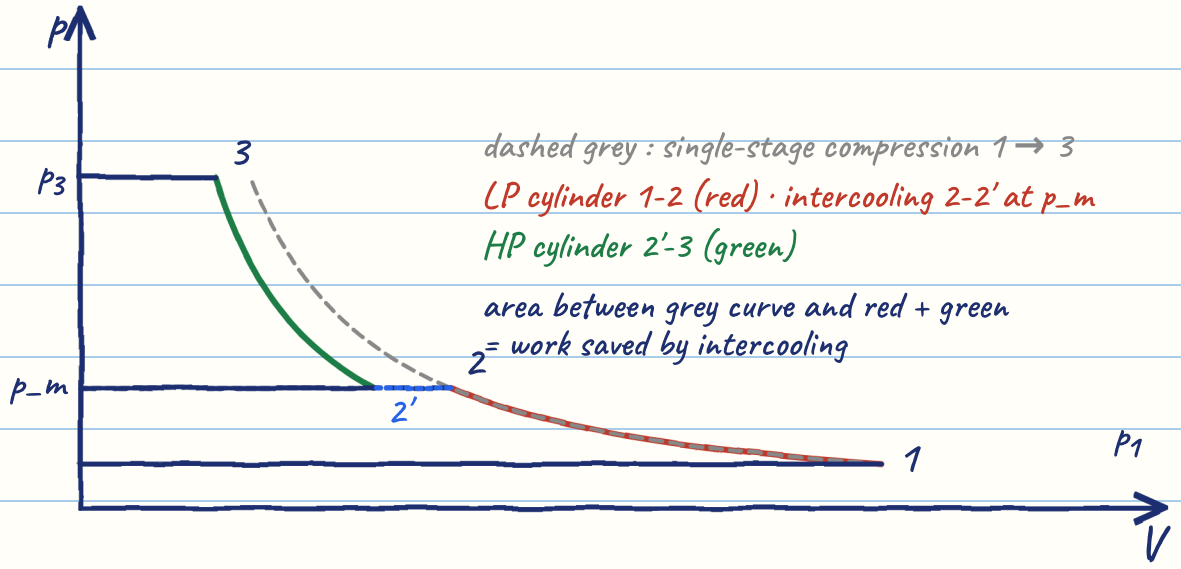
5.6 Multi-stage Compression – Why ?

- For high pressures (> 8 bar) a single stage is bad because : (1) **delivery temperature becomes very high** ($T_2 = T_1\, r_p^{(n-1)/n} \approx 300$ °C at $r_p = 16$) → lubricating oil flash point (≈ 200 °C) & valve problems, (2) **η_v falls** (clearance effect), (3) **more work** (poor cooling), (4) large cylinder size & heavy construction for high pressure / strong flywheel, (5) uneven torque.
- **Multi-stage** : compress in 2 or more cylinders with **intercoolers** between them. Advantages : saving of work, better η_v, lower delivery temperature, lighter cylinders (smaller HP cylinder), better balancing & more uniform torque. Disadvantage: more cost, space.



Perfect intercooling : air leaves intercooler at the original temperature T_1 ($T_3' = T_1$).
Minimum work for $p_m = \sqrt{(p_1\, p_3)}$ → equal pressure ratio in each stage.

5.7 Two-stage Compressor with Intercooling



- LP cylinder compresses 1→2 ($p_1 \to p_m$). The air passes through an **intercooler** at constant pressure p_m cooled 2→2' back to the initial temperature (**perfect intercooling**, $T_{2'} = T_1$, hence $p_m V_{2'} = p_1 V_1$). HP cylinder then compresses 2'→3 ($p_m \to p_3$).
- **Total work** (perfect intercooling, same n) :

$$W = [n/(n-1)]\, p_1 V_1 \left[(p_m/p_1)^{(n-1)/n} + (p_3/p_m)^{(n-1)/n} - 2 \right]$$

- **Optimum intermediate pressure** ($dW/dp_m = 0$) :

$$p_m = \sqrt{(p_1 p_3)} \Rightarrow p_m/p_1 = p_3/p_m$$

- Minimum work : $W_{min} = [2n/(n-1)]\, p_1 V_1 \left[(p_3/p_1)^{(n-1)/2n} - 1 \right]$
- For **z stages** : pressure ratio per stage $r = (p_{final}/p_1)^{1/z}$; $W_{min} = z\, n/(n-1)\, p_1 V_1 \left(r^{(n-1)/n} - 1\right)$. Work in each stage is equal ⇒ **equal work per stage**.
- **Heat rejected in the intercooler** per kg = $c_p (T_2 - T_1) = c_p T_1 \left(r^{(n-1)/n} - 1\right)$. Number of stages is chosen so that the delivery temperature per stage < 150-180 °C (usually 2 stages up to 30 bar, 3 up to 100 bar).

- **Interstage area** : cylinder volume ratio (LP : HP) = r (for equal ratios) → HP cylinder diameter smaller.

5.8 Effect of Pressure Drop & Imperfect Intercooling

- **Imperfect intercooling** : $T_{2'} > T_1 \Rightarrow p_m V_{2'} > p_1 V_1 \Rightarrow$ HP cylinder handles more volume ⇒ **more HP work**, saving is reduced; p_m shifts slightly.
- **Pressure drop in the intercooler / pipes / valves** : the HP cylinder draws air at $p_m' <$ LP delivery pressure $p_m \Rightarrow$ the HP suction line lies lower & LP delivery higher ⇒ **extra loop area (loss of work)**, lower η_v, higher power per m^3 of FAD. Keep intercooler $\Delta p < 0.1$-0.2 bar by large flow area.
- Also leakage in valves & rings, heating of suction air, and delivery valve resistance - increase the indicated work & reduce capacity.

Worked example 2 (two-stage)

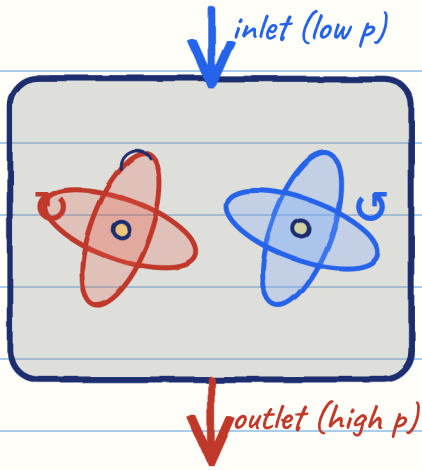
- A two-stage compressor delivers FAD 3 m^3/min from 1 bar, 27 °C to 16 bar, n = 1.3, perfect intercooling, optimum p_m. Find p_m, power & saving over single stage.
  - → $p_m = \sqrt{(1 \times 16)} =$ **4 bar** ; $p_1 V_1 = 10^5 \times 3/60 = 5000$ W
  - → $W_2 = (2 \times 1.3/0.3) \times 5000 \times [16^{0.3/2.6} - 1] = 8.667 \times 5000 \times 0.377 =$ **16.34 kW**
  - → Single stage : $W_1 = 4.333 \times 5000 \times [16^{0.3/1.3} - 1] = 21667 \times 0.896 =$ **19.42 kW** ⇒ saving ≈ **3.08 kW (≈ 16 %)**
  - → Delivery temp.: single stage 296 °C vs 140 °C per stage in two-stage ✓

5.9 Problems Related to Reciprocating Compressors

Problem	Cause / remedy
Valve leakage / failure	dirt, broken springs, carbon ⇒ filter, clean & replace valves
High discharge temperature	poor cooling, fouled intercooler ⇒ clean, more stages
Carbon deposits, fire / explosion	oil oxidation at high T near valves ⇒ right oil, cooling
Water carry-over	moist air condensed in intercooler ⇒ drain traps, aftercooler
Pulsating flow & vibration	intermittent delivery ⇒ air receiver, balancing, dampers
Ring & liner wear, oil carry-over	dust & poor lubrication ⇒ filter, oil separator
Knocking	liquid/water slugging, loose parts, excess clearance
Overheating of cylinders	low coolant flow ⇒ check pump, fins, jackets
Capacity control	unloaders, clearance pockets, speed variation, start-stop

- Safety valve on receiver & intercooler; pressure gauge; drain cocks; relief valves between stages; fusible plug for fire.

5.10 Rotary Compressors – Roots Blower



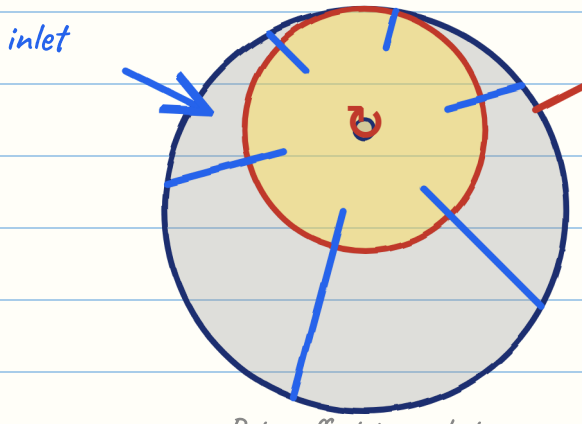
Two 2-lobe (figure-8) rotors, synchronised by timing gears, turn in opposite directions.

Air trapped between lobe & casing is carried to the delivery side - no internal compression (volume constant).

Pressure rises when delivery side back-flows.

- **Working principle** : two figure-of-eight (**two or three lobe**) rotors synchronised by timing gears rotate in opposite directions inside a casing with small (≈ 0.1 mm) clearances. Air is trapped between lobe & casing & **carried round** to the delivery side; there is **no internal compression** - pressure rises when the delivery is closed (back-flow compresses it). Delivery is pulsating.
- Features : simple, oil-free air (no contact), moderate pressure ratio (≤ 2, up to 0.5-1 bar), efficiency 50-70 %; used as supercharger, scavenging blower for 2-stroke diesels, vacuum pumps, pneumatic conveying, ventilation.
- Capacity ∝ speed (nearly constant volume machine).

5.11 Vane Type Blower (Sliding-vane compressor)



Rotor offset towards top

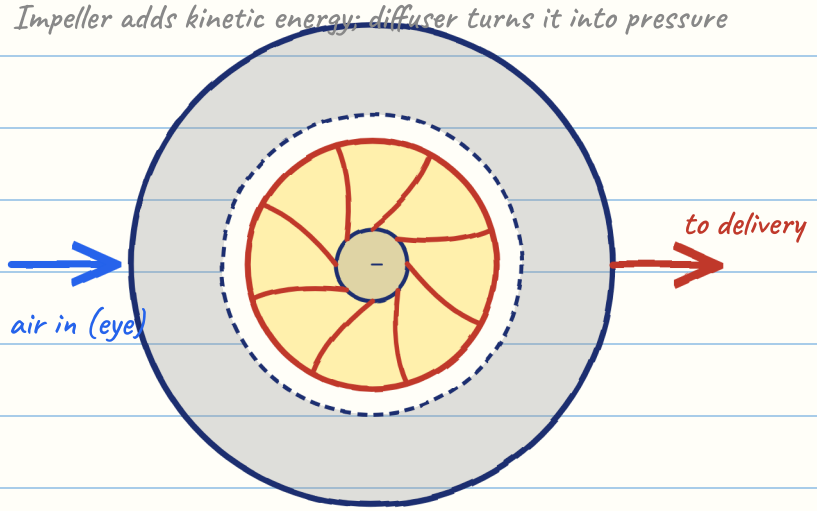
Eccentric rotor with sliding vanes in radial slots; centrifugal force presses vanes on the casing.

Trapped volume between vanes first increases (suction) then decreases (internal compression) → delivery.

- **Working** : a cylindrical rotor is mounted eccentrically in a cylindrical casing; **vanes** slide in radial slots & are thrown against the casing wall by centrifugal force. As the rotor turns the volume between consecutive vanes **increases at the suction port (suction)** and then **decreases (internal compression)** before the delivery port.
- Features : compact, smooth flow, oil-lubricated (sealed by oil), p-ratio up to 4-5 in single stage (8 for two-stage); vane wear & frictional heating (oil cooled). Used in refrigeration & air-conditioning compressors, vacuum pumps, small air supply, car power-steering, fuel pumps.

5.12 Centrifugal Compressor

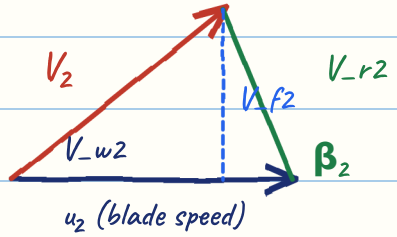
Impeller adds kinetic energy; diffuser turns it into pressure



Impeller (red) + vaneless / vaned diffuser (dashed) + volute casing

$W = u_2 V_{w2}$ (J/kg)

Outlet velocity triangle



- **Parts** : inlet casing, **impeller** (rotating blades, 3000-30000+ rpm), **diffuser** (vaneless / vaned) and **volute** (scroll) casing. Air enters axially at the eye, **centrifugal action** throws it outward ⇒ both **pressure & velocity** rise in the impeller; diffuser converts high velocity (kinetic energy) into pressure; volute collects & delivers.
- **Euler work** (per kg) : $W = u_2 V_{w2} - u_1 V_{w1}$; for axial entry $V_{w1} = 0$:

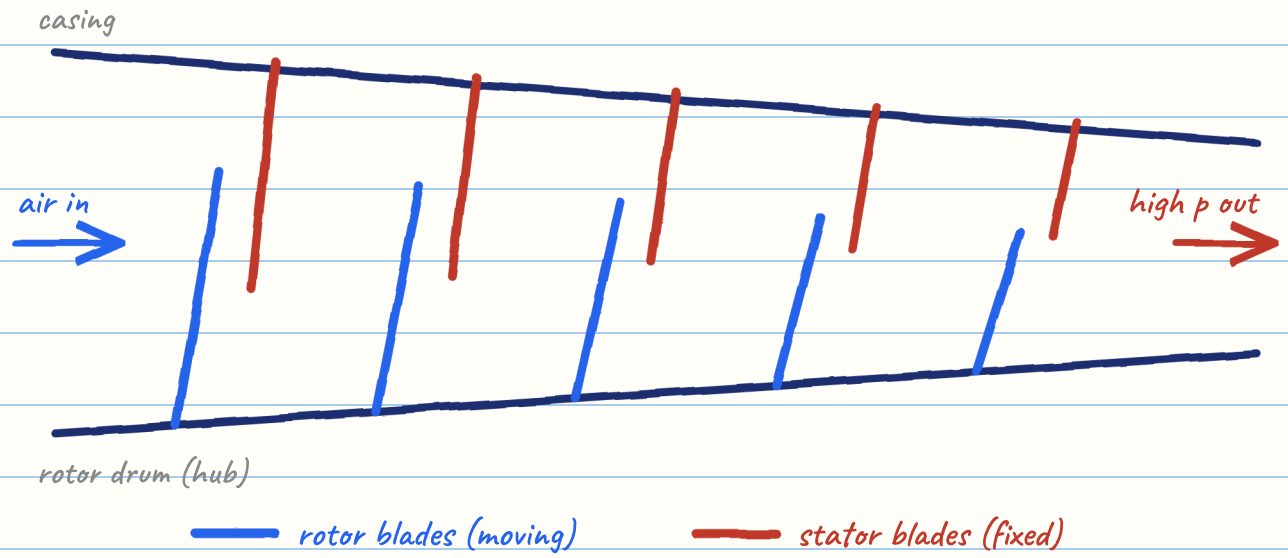
$$W = \sigma \psi u_2^2 = c_p (T_{02} - T_{01})$$ (σ = slip factor, ψ = work-input factor ≈ 1)

- **Slip** : the actual V_{w2} < ideal because the air lags behind the blade ⇒ slip factor $\sigma = V_{w2}/V_{w2}' \approx 0.85\text{-}0.9$.
- **Pressure ratio** per stage 3 - 5 (up to 8-10 with titanium impeller); isentropic efficiency 75-85 %.

$$p_{02}/p_{01} = [1 + \eta_c \psi \sigma u_2^2 / (c_p T_{01})]^{\gamma/(\gamma-1)}$$

- **Limits** : **surge** (flow reversal at low flow ⇒ violent pulsation), **choking** (Mach 1 at the inducer / diffuser throat at high flow); the **operating range** lies between surge line & choke line.
- Uses : gas-turbine & turbocharger compressors (small), large process / blast-furnace air, refrigeration (chillers), air-conditioning. Suited for large flow rates, compact, oil-free, steady delivery.

5.13 Axial Flow Compressor



- **Principle** : air flows **parallel to the axis** through alternating rows of **rotor blades** (give kinetic energy / raise velocity & some pressure) & **stator blades** (diffuse: convert velocity into pressure and guide air to the next rotor). One rotor row + one stator row = **one stage**; annulus area decreases to keep axial velocity constant as density rises.
- Typical stage pressure ratio only **1.1 - 1.4** (ΔT 15-30 K) ⇒ many stages (8 to 20) are used for an overall ratio 10-40.

- **Stage work** : $W = u\,(V_{w2} - V_{w1}) = c_p \Delta T_{0s}$; degree of reaction $R \approx 0.5$ (symmetrical blading); **de Haller number** ($V_2/V_1 > 0.72$) limits diffusion to avoid blade stall.
- **Stall & surge** : rotating stall (cells of separation) and surge limit the stable operating range. Isentropic efficiency 85-92 %.
- Uses : **jet engines & gas turbines**, power-plant, wind tunnels, blowers.

5.14 Comparison of Compressors

Point	Reciprocating	Roots	Vane	Centrifugal	Axial
Type	positive	positive	positive	dynamic	dynamic
Flow	small	medium	small-med.	large	very large
Pressure ratio	very high	≤ 2	4 - 8	3 - 10	10 - 40 (multi-stage)
Delivery	pulsating	pulsating	smooth	steady	steady
Efficiency	70-85 %	50-70 %	60-75 %	75-85 %	85-92 %
Speed	low	medium	medium	very high	very high
Size / cost	big, heavy	small	small	compact	compact, costly

Important questions (exam)

- Derive work of single-stage reciprocating compressor : isothermal & polytropic.
- Derive volumetric efficiency in terms of clearance ratio & pressure ratio.
- Derive the condition for minimum work in two-stage compressor with intercooler; show $p_m = \sqrt{(p_1 p_2)}$.

- Effect of pressure drop & imperfect intercooling in multistage compressors.
- Working of Roots blower, vane blower, centrifugal & axial-flow compressors (neat sketches).